

Diagonal Mellin Reassembly on Actual Orbit Fibers: Scalar Saturation, Profile Overlap, and an Interior TPC Bridge

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

The preceding twin-prime-correlation reduction proved stable sampled frames for fixed scalar Mellin families but left open their transfer to the actual, growing family of row-dependent orbit translates. We show that two different questions were hidden in that gap. If the orbit label is discarded, a cluster of $r + 1$ normalized scalar translates has Gram eigenvalue at most $O_r(J^{-2r})$. Thus the unreduced scalar family has no uniform lower frame, and on an interior translate-independent window its exact dimension is of order J , so no polylogarithmic scalar bank can preserve the energy of every vector in that exact span with uniform relative error.

The physical coefficientwise space does not discard the orbit label. It retains (m, r, j, γ) as orthogonal coordinates before coherent prime summation and residual grouping. On those coordinates we prove an exact q -factor diagonal Mellin identity. For fixed $W_1, \dots, W_q \in C_c^\infty((0, \infty))$, base coefficients c_i , and positive samples $x_{i,\nu}$, its weighted canonical envelope and physical output satisfy

$$\mathcal{E}_B(\mathbf{W}; c) = A_B(\mathbf{W})\|c\|_2, \quad \|\mathcal{S}\mathbf{W}c\|_2^2 = \sum_i |c_i|^2 \prod_{\nu=1}^q |W_\nu(x_{i,\nu})|^2.$$

Consequently the actual-direction condition number is exactly

$$\frac{\mathcal{E}_B(\mathbf{W}; c)^2}{\|\mathcal{S}\mathbf{W}c\|_2^2} = \frac{A_B(\mathbf{W})^2}{\Theta(\mathbf{W}; c)},$$

where Θ is the nonnegative physical profile-to-carrier ratio. This bypasses a lower-frame assertion for arbitrary Mellin densities or for the entire growing scalar family.

We then give a conditional criterion for evaluating its sampled actual instantiation Θ_X^{act} on a literal prime-squarefree row universe. The declared factors include $W(c_\beta e^{x+yt})$ and $\Psi(c'_\beta e^{yt})$, with the normalized source parameter β in a fixed compact set. If the continuous profile-to-carrier ratio has a uniform lower bound $\rho_* > 0$, and if the displayed profiles and weights satisfy an exact coordinatewise dictionary with one complete nondegenerate row cell, fixed-modulus row sampling, orbit Riemann sampling, and the literal-mask degree estimate give

$$\Theta_X^{\text{act}} \geq \rho_*/2$$

uniformly for arbitrary source concentration and arbitrary finite retained cofactor support. Hence the canonical coefficientwise pre-terminal direction has $O(1)$ condition number on every such uniformly active direct cell. Under the same exact dictionary and nondegeneracy hypotheses, we verify the required regularity-to-carrier normalization and projected-reference lower bound and deduce polylogarithmic row-log Fourier compressibility of the retained Hilbert-valued orbit field; this does not compress the orbit labels.

The overlap hypothesis is essential. Two fixed smooth bumps whose scaled supports graze can make Θ_X^{act} smaller than every power of X , while the canonical layer envelope remains of carrier size. The resulting super-polynomial near-kernel concerns the distinguished actual direction, not an arbitrary signed density. Thus the paper proves an exact diagonal identity and a conditional coefficientwise Mellin bridge on uniformly active interior packets. Support grazing is a sharp obstruction to the overlap route, but the paper does not verify the exact-profile dictionary or uniform overlap for every inherited packet. Coherent prime summation, residual grouping, localization to the physical shift, native-Gram identification, and every parity-sensitive conclusion remain open.

Keywords: Mellin inversion; orbit fibers; profile overlap; scalar Gram saturation; Hilbert-valued sampling; prime-squarefree rows; static mask; sieve parity.

Convention. All Hilbert spaces are complex. Every Mellin contour is the real axis. Direct residue cells are formed before the literal static mask, and the mask is formed before any signed projective representation. Constants may depend on the fixed smooth shapes, residue moduli, and derivative order, but not on X . We write $e^z := \exp z$, $e(z) := \exp(2\pi iz)$, and $e_q(z) := e(z/q)$.

MSC 2020: Primary 42A38, 46C05; Secondary 11N35, 42B35, 15A18.

1 Introduction

The smooth row-orbit factors in the opened fixed-shift packet are not a fixed scalar family. They contain expressions such as

$$W(mj/X), \quad \Psi(dj/K), \quad (1)$$

and therefore generate more and more log translates as the orbit length J grows. TPC-51 proved that distinct compactly supported log translates are linearly independent and consequently cannot be replaced by a fixed-rank scalar family merely by relabelling [6]. It also proved stable sampled quotient frames for every declared fixed finite family and for polylogarithmic row-log Fourier banks. The missing step was to connect those model results to the actual coefficientwise operator.

There are two traps at this point. First, linear independence is not a uniform lower-frame theorem. Nearby translates are almost parallel, and their scalar Gram matrix is extremely ill-conditioned. Second, the physical operator never forms that scalar Gram: before coherent prime summation it retains the orbit coordinate j , together with source, cofactor, opposite-row, and residual labels. Those labels are orthogonal. Forgetting them manufactures cross terms that are absent from the physical coefficient space.

This paper separates the two issues. We give a quantitative saturation theorem for the false scalar collapse, then prove a diagonal theorem for the label-preserving actual direction. The latter does not assert a lower frame on every signed Mellin density. It uses the one coefficient density fixed by provenance and reduces its condition number to a nonnegative physical overlap ratio.

1.1 Main results

The conclusions are organized by the operator on which they hold.

- (R1) **Scalar saturation.** Let $a \mapsto g(a)$ be a normalized C^∞ Hilbert-valued curve. If $r + 1$ sampling parameters lie in an interval of length $O_r(J^{-1})$, then their scalar Gram matrix has

$$\lambda_{\min} \leq C_r J^{-2r}. \quad (2)$$

For interior log translates with no exact relation, this persists on the exact span. Their dimension is of order J , so an exact or uniform-relative scalar reduction needs rank of order J , not a fixed or polylogarithmic rank.

- (R2) **Orthogonal orbit lift.** For arbitrary vectors f_j , the label-preserving synthesis $c \mapsto \bigoplus_j c_j f_j e_j$ has diagonal Gram

$$\left\| \bigoplus_j c_j f_j e_j \right\|^2 = \sum_j |c_j|^2 \|f_j\|^2. \quad (3)$$

Thus scalar saturation is not a physical obstruction while j remains an output coordinate.

- (R3) **Exact multi-factor Mellin ledger.** For one direct residue cell and one provenance-fixed coefficient vector, multidimensional Mellin inversion gives

$$\frac{\mathcal{E}_B(\mathbf{W}; c)^2}{\|\mathbf{S}\mathbf{W}c\|_2^2} = \frac{A_B(\mathbf{W})^2}{\Theta(\mathbf{W}; c)}. \quad (4)$$

The numerator contains only the fixed weighted Wiener–Mellin mass; all scale dependence is in the explicit profile-overlap Θ . This is an identity, not a numerical conditioning heuristic.

- (R4) **Uniformly active actual cells.** Normalize an opposite row by $x = \log(\ell/L)$, $y = \log(d/D)$, and the orbit by $t = j/J$. For a source-scale parameter β , let B_β be the carrier and F_β the unseparated physical profile. Define

$$\rho_* := \inf_{\beta \in \mathfrak{B}} \frac{\int |F_\beta(x, y, t)|^2 e^{x+y} dx dy dt}{\int |B_\beta(x, y, t)|^2 e^{x+y} dx dy dt}. \quad (5)$$

A single strict overlap point yields a nonempty compact uniformly active subcell; on such a cell, $\rho_* > 0$. Conditional on the named packet also satisfying the exact actual-to-profile coordinate dictionary and complete-row hypotheses, the prime-squarefree row sampler, the orbit Riemann sum, and the literal mask imply $\Theta_X^{\text{act}} \geq \rho_*/2$, uniformly for arbitrary source concentration and arbitrary finite retained cofactor support. Hence that declared coefficientwise pre-terminal direction has condition number $O(1)$. The paper does not infer this dictionary or uniform overlap for every inherited packet.

- (R5) **Conditional row-log compression.** A tensor cosine bank of degree K_F has squared approximation tail $O(K_F^{-2})$ for a retained Hilbert-valued orbit field with controlled row-log regularity. If that regularity is bounded by the carrier energy and the projected reference energy has a fixed lower bound, then $K_F = (\log X)^A$, arithmetic sampling, and sparse mask deletion give an $o(1)$ relative tail. Under the exact-profile dictionary and $\rho_* > 0$, the required normalizations are verified for the declared retained actual field. The projection compresses row-log frequency, not the orbit label.
- (R6) **Grazing is a sharp boundary.** Two fixed C_c^∞ bumps can have scaled supports whose overlap approaches a common flat endpoint. At distance $\delta_X = X^{-\sigma}$, the physical overlap is smaller than every power of X while the Mellin carrier stays nonzero. The distinguished actual direction then has a super-polynomial condition number. Thus bounded smoothness and dyadic scale comparability do not replace the positive-overlap hypothesis.

1.2 What advances and what does not

We use the same three-level ledger as the preceding reduction.

Level L0: exact algebra and boundary analysis.

Scalar saturation, the orthogonal-lift identity, the multi-factor Mellin ledger, and the grazing construction are exact analytic statements.

Level L1: physical coefficient interface.

Under the exact-profile and complete-row hypotheses, the overlap theorem keeps the literal source coefficient, direct residue cell, orbit label, opposite row, residual label, real-axis Mellin normalization, and static mask in their inherited order. It gives a conditional actual-direction Mellin bridge on uniformly active direct cells. Verifying those hypotheses for every inherited packet is not part of the theorem.

Level L2: arithmetic target.

A coherent prime square, fixed-shift target correlation, or parity estimate. No statement at this level is proved here.

The L1 conclusion is a strict subgate. It is not the full TPC-48 profile envelope: that object takes a Hilbert norm after coherent prime summation and before the outer signed projective integral. Nor does the present theorem group $s = d(m)r$, localize a frozen family to the physical h_0 , or identify the residual Gram with the native affine physical Gram. In particular, no fixed-shift prime-pair lower bound, Hardy–Littlewood asymptotic, parity-barrier breach, or twin-prime conclusion is asserted.

1.3 Organization

Section 2 reconstructs the exact coefficientwise operator. Section 3 proves scalar saturation and the orthogonal-lift identity. The diagonal Mellin ledger is section 4. The continuous and sampled overlap theorems are established in section 5, and their actual TPC consequence in section 6. Section 7 proves the corresponding conditional polylogarithmic relative-tail theorem. The sharp grazing obstruction and the endpoint claim ledger are given in sections 8 and 9. The exact finite certificate is described in section 10.

2 The retained-label actual coefficient interface

We reconstruct the operator before proving an estimate. The order of labels and norms is part of the theorem: a diagonal Mellin identity is available only before coherent prime summation and residual grouping. The interface is inherited from TPC-25, TPC-36, and TPC-47–51 [1–6].

2.1 Rows, the literal mask, and the actual profile

Fix the physical shift $h_0 \neq 0$ and one source-compatible dyadic packet. A row is

$$\rho = (\ell_\rho, d_\rho), \quad m_\rho = \ell_\rho d_\rho \asymp Q, \quad (6)$$

where ℓ_ρ is prime, d_ρ is squarefree, and the inherited size separation makes m_ρ determine ρ . Let $\mathcal{A}_1, \mathcal{A}_2$ be the source and opposite-row systems. The literal static mask is

$$K_{\alpha\gamma} := \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > \Delta} \mathbf{1}_{(d_\alpha, d_\gamma) \leq G}, \quad \Delta = QX^{-\kappa_0}, \quad G = X^{\kappa_0}. \quad (7)$$

It is not replaced by a signed rank-one representation in this section.

Before the mask is decomposed, the joint multiplier is

$$\mathfrak{A}_{m,\gamma}^{\text{act}}(j) = K_{m\gamma} \Xi_{m,\gamma}(j) \mathcal{W}_{m,\gamma}(j/J), \quad (8)$$

where Ξ has fixed period $H_0 = O_{h_0}(1)$, and $\mathcal{W}$ is the complete unseparated product of the declared smooth row-orbit factors. On the left, the literal source coefficient has the form

$$\gamma_m^{(1)} = \mu(d(m))(\log \ell(m))\omega_D(d(m))\psi_L(\ell(m)/L)\zeta_m^{(1)}. \quad (9)$$

The factor $\zeta_m^{(1)}$ is kept exactly; no lower bound or source equidistribution is assumed below.

Fix one complete source/opposite-row residue group of positive density and suppress its outer label. A fixed finite collection of such row groups is handled later by orthogonal summation. Let $\mathcal{A}_0$ be the direct orbit residue cells inside the chosen row group, let $\mathcal{O}_L$ be its common opposite-row set, and let $\mathcal{I}_X$ contain the supported pairs (m, r) before prime summation. We identify $a \in \mathcal{A}_0$ with its orbit residue modulo H_0 and put

$$\begin{aligned} \mathcal{V}_L &:= \ell^2(\mathcal{O}_L \times \mathcal{S}_X), \\ \mathcal{J}_{X,a} &:= \{j \in \mathbb{Z} : j/J \in I_0, j \equiv a \pmod{H_0}\}, \\ \mathcal{H}_X^{\text{coeff}} &:= \bigoplus_{a \in \mathcal{A}_0} \bigoplus_{(m,r) \in \mathcal{I}_X} \ell^2(\mathcal{J}_{X,a}; \mathcal{V}_L). \end{aligned} \quad (10)$$

where $I_0 \Subset (0, \infty)$ is a fixed common normalized orbit interval. All index sets are finite for each X .

For a direct residue cell $a \in \mathcal{A}_0$, the actual profile is

$$G_{m,r,a}^L(j) := \sum_{\gamma \in \mathcal{O}_L} \gamma_m^{(1)} K_{m\gamma} \Xi_{m,\gamma}(j) \mathcal{W}_{m,\gamma}(j/J) e_{\gamma,d(m)r} \in \mathcal{V}_L, \quad j \in \mathcal{J}_{X,a}. \quad (11)$$

Thus the coefficientwise physical vector is

$$G_X^L := \bigoplus_{a \in \mathcal{A}_0} \bigoplus_{(m,r) \in \mathcal{I}_X} (G_{m,r,a}^L(j))_{j \in \mathcal{J}_{X,a}} \in \mathcal{H}_X^{\text{coeff}}. \quad (12)$$

Because all displayed labels are orthogonal,

$$\begin{aligned} \|G_X^L\|_{\mathcal{H}_X^{\text{coeff}}}^2 &= \sum_{a \in \mathcal{A}_0} \sum_{(m,r) \in \mathcal{I}_X} \sum_{j \in \mathcal{J}_{X,a}} \sum_{\gamma \in \mathcal{O}_L} |\gamma_m^{(1)}|^2 K_{m\gamma} \\ &\quad \times |\Xi_{m,\gamma}(j)|^2 |\mathcal{W}_{m,\gamma}(j/J)|^2. \end{aligned} \quad (13)$$

There is no cross term between different (m, r, j, γ) . In particular, two pairs (m, r) that later produce the same residual label have not yet been grouped.

2.2 The canonical Mellin synthesis

Direct residue splitting and real-axis Mellin inversion give a fixed complex provenance measure λ_X on a finite disjoint union of spaces $F_a \times \mathbb{R}^{q_a}$. Write its polar decomposition as

$$d\lambda_X(\theta) = \varsigma_X(\theta) d\mu_X(\theta), \quad \mu_X = |\lambda_X|, \quad |\varsigma_X| = 1 \quad \mu_X\text{-a.e.} \quad (14)$$

For a direct cell, its literal-mask layer $k_{X,\theta}^L \in \mathcal{H}_X^{\text{coeff}}$ has coordinates

$$\begin{aligned} k_{X,\theta}^L(a; m, r, j) &:= u_{a,\tau}^L(m) \phi_{a,\tau}(j/J) \\ &\quad \times \sum_{\gamma \in \mathcal{O}_L} K_{m\gamma} v_{a,\tau}(\gamma) e_{\gamma,d(m)r}. \end{aligned} \quad (15)$$

No projective-mask factor $p_\eta q_\eta$ occurs in (15). Define

$$A_{X,\theta}^L := \varsigma_X(\theta) k_{X,\theta}^L, \quad \mathcal{S}_X^L f := \int f(\theta) A_{X,\theta}^L d\mu_X(\theta). \quad (16)$$

The transform densities are Schwartz, while the finite-dimensional layer norm has at most polynomial growth in the Mellin variables. Hence its norm is integrable against the transform variation, and (16) is a genuine Bochner integral on its natural domain.

The provenance-fixed direction is $f = \mathbf{1}$. Coordinatewise Mellin inversion gives the exact identity

$$\boxed{\mathcal{S}_X^L \mathbf{1} = G_X^L.} \quad (17)$$

For a fixed polynomial profile weight $w_B(\theta) \geq 1$, the displayed canonical envelope is

$$\mathcal{E}_{1,B}^L(f) := \int w_B(\theta) |f(\theta)| \|A_{X,\theta}^L\|_{\mathcal{H}_X^{\text{coeff}}} d\mu_X(\theta), \quad (18)$$

and the actual-direction condition number is

$$\kappa_{\text{can},B}^L(X) := \frac{\mathcal{E}_{1,B}^L(\mathbf{1})^2}{\|G_X^L\|_{\mathcal{H}_X^{\text{coeff}}}^2}, \quad (19)$$

when $G_X^L \neq 0$.

2.3 Normalized profile geometry

The original provenance contains fixed smooth factors such as $W(m_\gamma j/X)$ and $\Psi(d_\gamma j/K)$ [1]. Put

$$x_\gamma = \log(\ell_\gamma/L), \quad y_\gamma = \log(d_\gamma/D), \quad t_j = j/J. \quad (20)$$

Then

$$\frac{m_\gamma j}{X} = \frac{LDJ}{X} e^{x_\gamma + y_\gamma t_j}, \quad \frac{d_\gamma j}{K} = \frac{DJ}{K} e^{y_\gamma t_j}. \quad (21)$$

The packet relations $LDJ \asymp X$ and $DJ \asymp K$ keep the two leading scale factors in fixed compact subsets of $(0, \infty)$.

After one of finitely many direct residue and smooth subdivisions, we write the modulus of the unseparated profile as

$$F_\beta(x, y, t) = R_\beta(x, y, t) \prod_{\nu=1}^q W_\nu(z_{\nu,\beta}(x, y, t)), \quad \beta \in \mathfrak{B}, \quad (22)$$

where:

- (a) $\mathfrak{B}$ is a fixed compact set of normalized source-scale parameters;
- (b) R_β is the carrier formed from the remaining fixed cutoffs and the direct residue amplitude;
- (c) each $W_\nu \in C_c^\infty((0, \infty))$ is a declared physical shape; and
- (d) the maps $z_{\nu,\beta}$ and all displayed profiles form a uniformly bounded C^∞ family on

$$\Omega := [0, \log 2]^2 \times I_0. \quad (23)$$

Typical arguments are

$$z_{1,\beta}(x, y, t) = c_\beta e^{x+y} t, \quad z_{2,\beta}(x, y, t) = c'_\beta e^y t, \quad (24)$$

with c_β, c'_β in compact positive intervals. Formula (22) is an assumption on the declared direct packet, not an assertion that the two examples in (24) exhaust every inherited smooth factor.

2.4 The downstream boundary

Only after (12) is formed does the terminal map introduce primes and the determinant slice:

$$\mathcal{F}_{h_0}^{\text{act,L}} = \sum_{\substack{a,p,r,m,j: pr-mj=h_0 \\ (m,r) \in \mathcal{I}_X, j \in \mathcal{J}_{X,a}}} \mu(r) \Lambda(pr) G_{m,r,a}^L(j) \otimes e_j. \quad (25)$$

At fixed (m, r) , the p -sum is coherent. The present paper neither moves (18) through this map nor reverses the resulting Minkowski inequality. Likewise, the maps grouping $s = d(m)r$, completing frozen shifts, and identifying the native affine Gram all occur downstream.

3 Scalar saturation and the orthogonal orbit lift

This section separates two operators that have very different lower-frame behaviour. The first forgets the orbit label and adds a growing collection of nearby scalar profiles in one Hilbert space. The second retains that label as an orthogonal output coordinate. The saturation theorem below concerns the first, unreduced Gram matrix. It is not, by itself, a statement about the smallest positive eigenvalue after quotienting exact relations.

3.1 A clustered smooth curve has a small Gram eigenvalue

Let $\mathbf{H}$ be a complex Hilbert space. For vectors $v_0, \dots, v_r \in \mathbf{H}$, we use the Gram convention

$$\text{Gram}(v_0, \dots, v_r)_{k\ell} := \langle v_\ell, v_k \rangle_{\mathbf{H}}. \quad (26)$$

Thus, for every $c \in \mathbb{C}^{r+1}$,

$$c^* \text{Gram}(v_0, \dots, v_r) c = \left\| \sum_{\ell=0}^r c_\ell v_\ell \right\|_{\mathbf{H}}^2. \quad (27)$$

Theorem 3.1 (Clustered-curve saturation). *Let $I \subset \mathbb{R}$ be an interval, let $r \geq 1$, and let $g \in C^r(I; \mathbf{H})$ satisfy $\|g(a)\|_{\mathbf{H}} = 1$ on I . Fix a compact interval $I_1 \Subset I$. Suppose that, for some $J \geq 1$, there are $r+1$ distinct points $a_0, \dots, a_r \in I_1$ and a point $a_* \in I_1$ such that*

$$\max_{0 \leq \ell \leq r} |a_\ell - a_*| \leq \frac{D}{J} \quad (28)$$

for a fixed $D > 0$. Then

$$\lambda_{\min}(\text{Gram}(g(a_0), \dots, g(a_r))) \leq \frac{(r+1)D^{2r}}{(r!)^2 J^{2r}} \sup_{a \in I_1} \|g^{(r)}(a)\|_{\mathbf{H}}^2. \quad (29)$$

Consequently, the least eigenvalue of any larger normalized Gram matrix containing these vectors is bounded by the right-hand side of (29).

Proof. The homogeneous system

$$\sum_{\ell=0}^r c_\ell (a_\ell - a_*)^q = 0, \quad 0 \leq q < r, \quad (30)$$

has r equations and $r+1$ unknowns. Choose a nonzero solution and normalize it by $\sum_\ell |c_\ell|^2 = 1$. Taylor's formula with integral remainder gives

$$g(a_\ell) = \sum_{q=0}^{r-1} \frac{g^{(q)}(a_*)}{q!} (a_\ell - a_*)^q + R_\ell, \quad \|R_\ell\|_{\mathbf{H}} \leq \frac{|a_\ell - a_*|^r}{r!} \sup_{a \in I_1} \|g^{(r)}(a)\|_{\mathbf{H}}. \quad (31)$$

The line segment between a_* and every a_ℓ lies in I_1 , because I_1 is an interval. The polynomial terms in (31) cancel by (30). Hence Cauchy–Schwarz and (28) give

$$\begin{aligned} \left\| \sum_{\ell=0}^r c_\ell g(a_\ell) \right\|_{\mathbf{H}} &\leq \sum_{\ell=0}^r |c_\ell| \|R_\ell\|_{\mathbf{H}} \\ &\leq \frac{\sqrt{r+1}D^r}{r!J^r} \sup_{a \in I_1} \|g^{(r)}(a)\|_{\mathbf{H}}. \end{aligned} \quad (32)$$

The Rayleigh principle and (27) prove (29). The assertion for a larger Gram matrix follows either by extending c by zero or by Cauchy interlacing. $\square$

The normalization in the theorem costs no regularity on an interior nonzero family.

Corollary 3.2 (Normalization and the logarithmic orbit grid). *Let $f \in C^r(I; \mathbf{H})$, and suppose that*

$$\inf_{a \in I_1} \|f(a)\|_{\mathbf{H}} > 0 \quad (33)$$

on a compact interval $I_1 \subseteq I$. Then $g(a) := f(a)/\|f(a)\|_{\mathbf{H}}$ is a C^r normalized curve on I_1 , and the conclusion of theorem 3.1 holds with a constant depending only on r, D , the lower bound in (33), and the C^r -bounds for f .

In particular, fix $0 < z_- < z_+ < \infty$, a residue modulus $H_0 \geq 1$, and put

$$a_j := \log(j/J), \quad z_-J \leq j \leq z_+J, \quad j \equiv a \pmod{H_0}. \quad (34)$$

For every fixed r , any $r+1$ consecutive nodes in this progression form a cluster of width $O_{r,H_0,z_-}(J^{-1})$. Therefore their normalized scalar Gram has least eigenvalue

$$\lambda_{\min} \ll_{r,H_0,z_-,f} J^{-2r}. \quad (35)$$

At the inherited scale $J = X^{133/400+o(1)}$, the case $r=1$ already gives a condition-number lower bound $X^{133/200+o(1)}$ whenever the selected two-vector Gram is nonsingular. For every fixed r , the corresponding lower bound is $X^{133r/200+o(1)}$.

Proof. The norm map is C^r away from zero, so the first assertion follows from the chain and quotient rules. For the second, if $j \asymp J$, then

$$\log \frac{j+H_0}{j} = \log \left(1 + \frac{H_0}{j} \right) = O_{H_0,z_-}(J^{-1}). \quad (36)$$

Summing at most r such spacings verifies (28). Apply theorem 3.1. A normalized Gram has diagonal entries one and hence largest eigenvalue at least one. Dividing by (35) gives the stated condition-number bounds when the least eigenvalue is positive. $\square$

Remark 3.3 (Unreduced Gram versus exact quotient). The bounds above concern the least eigenvalue of the unreduced coefficient Gram. If the selected vectors have an exact relation, that least eigenvalue is zero, which is an even stronger failure of a full-space lower frame. Quotienting by the exact synthesis kernel removes zero eigenvalues. A bound for the smallest *positive* quotient eigenvalue therefore requires an additional verification that the selected profiles have no exact relation. The translate theorem below supplies that verification on one important interior model; no such verification is inferred merely from smooth dependence on the parameter.

3.2 Exact scalar rank and metric entropy

Proposition 3.4 (Interior translates: rank and scalar-bank lower bound). *Let $0 \neq w \in C_c^\infty(\mathbb{R})$, let $a_1, \dots, a_M \in \mathbb{R}$ be distinct, and put $w_j(s) := w(s + a_j)$. Then the w_j are linearly independent in $L^2(\mathbb{R})$. If an open interval U contains $\bigcup_{j=1}^M \text{supp } w_j$, their restrictions are also linearly independent in $L^2(U)$.*

Let $V_M := \text{span}\{w_1|_U, \dots, w_M|_U\}$, so that $\dim_{\mathbb{C}} V_M = M$. If $B \subset L^2(U)$ is a linear space and a linear map $P : V_M \rightarrow B$ satisfies

$$\|v - Pv\|_{L^2(U)} \leq \eta \|v\|_{L^2(U)} \quad (v \in V_M) \quad (37)$$

for some $0 \leq \eta < 1$, then

$$\dim_{\mathbb{C}} B \geq M. \quad (38)$$

Moreover, if $\mathbb{B}_{V_M}$ is the unit ball of V_M , then for $0 < \eta < 1$ its covering number in the $L^2(U)$ metric obeys

$$\mathcal{N}(\mathbb{B}_{V_M}, \eta) \geq \eta^{-2M}. \quad (39)$$

Proof. If $\sum_j c_j w(s + a_j) = 0$ on $\mathbb{R}$, Fourier transformation gives

$$\widehat{w}(\xi) \sum_{j=1}^M c_j e^{ia_j \xi} = 0. \quad (40)$$

The entire function $\widehat{w}$ is not identically zero, and hence is nonzero on some real open interval. The exponential polynomial in (40) vanishes on that interval, so it vanishes identically. Its first M derivatives at one point form a Vandermonde system in the distinct a_j , forcing every $c_j = 0$. If the restrictions have a relation on U , the linear combination is supported in U and vanishes there, hence vanishes on all of $\mathbb{R}$. This proves the rank assertion.

If $Pv = 0$, (37) gives $\|v\| \leq \eta \|v\|$, so $v = 0$. Thus P is injective on the M -dimensional space V_M , proving (38). Finally regard V_M as a real Euclidean space of dimension $2M$. Orthogonally projecting the centres of any radius- η cover onto V_M does not increase distances. Comparing $2M$ -dimensional volumes of the unit ball and radius- η balls yields (39). $\square$

Thus, whenever an interior scalar translate family has $M \asymp J$, every scalar bank satisfying a uniform relative-output estimate with error below one has rank $\gg J$. At $J = X^{133/400+o(1)}$, a scalar bank of polylogarithmic rank cannot do this. This rank statement is conditional on the displayed interior translate model; it is not being promoted to a quotient-rank theorem for every restricted product profile.

3.3 Retaining the orbit coordinate

The physical coefficient space uses a different synthesis.

Theorem 3.5 (Exact orthogonal-lift identity). *Let (Ω, σ) be a measure space, let $\mathbb{K}$ be a complex Hilbert space, and let S be a finite set. For $f_j \in L^2(\sigma; \mathbb{K})$, define*

$$\mathbb{T}^\perp c(x) := \bigoplus_{j \in S} c_j f_j(x) \in \ell^2(S; \mathbb{K}), \quad c = (c_j)_{j \in S} \in \mathbb{C}^S. \quad (41)$$

Then

$$\|\mathbb{T}^\perp c\|_{L^2(\sigma; \ell^2(S; \mathbb{K}))}^2 = \sum_{j \in S} |c_j|^2 \|f_j\|_{L^2(\sigma; \mathbb{K})}^2. \quad (42)$$

Consequently its coefficient Gram is the diagonal matrix

$$\text{diag} \left(\|f_j\|_{L^2(\sigma; \mathbb{K})}^2 : j \in S \right). \quad (43)$$

After discarding the zero vectors and normalizing each nonzero f_j , the Gram is exactly the identity. All statements remain true when σ is a finite weighted sampling measure.

Proof. For almost every x , orthogonality of the coordinate copies of $\mathbb{K}$ gives

$$\left\| \bigoplus_{j \in S} c_j f_j(x) \right\|^2 = \sum_{j \in S} |c_j|^2 \|f_j(x)\|_{\mathbb{K}}^2. \quad (44)$$

Integrate the finite nonnegative sum. The Gram and normalization statements follow immediately. $\square$

The theorem does not reduce the number of orbit coordinates. It says that a row-log frequency bank may have Hilbert-valued coefficients carrying all those coordinates without creating the scalar cross terms from [theorem 3.1](#).

Proposition 3.6 (Sparse deletion through leverage). *Let $\mathcal{R}$ be a finite row set of size N , let $w_\gamma > 0$, and let $M(\alpha, \gamma) \in \{0, 1\}$ satisfy*

$$\#\{\gamma \in \mathcal{R} : M(\alpha, \gamma) = 0\} \leq \varepsilon N \quad (45)$$

for every source label α . For a finite orbit set S , let $f_{\alpha,j} : \mathcal{R} \rightarrow \mathbb{K}$, and suppose

$$E_{\alpha,j} := \sum_{\gamma \in \mathcal{R}} w_\gamma \|f_{\alpha,j}(\gamma)\|_{\mathbb{K}}^2 > 0. \quad (46)$$

Define the leverage

$$\mathfrak{L}_{\alpha,j} := \frac{N \max_{\gamma \in \mathcal{R}} w_\gamma \|f_{\alpha,j}(\gamma)\|_{\mathbb{K}}^2}{E_{\alpha,j}}, \quad \mathfrak{L}_* := \sup_{\alpha,j} \mathfrak{L}_{\alpha,j}. \quad (47)$$

Then, for every $c \in \mathbb{C}^S$,

$$\begin{aligned} \sum_{j \in S} |c_j|^2 \sum_{\gamma \in \mathcal{R}} M(\alpha, \gamma) w_\gamma \|f_{\alpha,j}(\gamma)\|_{\mathbb{K}}^2 \\ \geq (1 - \varepsilon \mathfrak{L}_*) \sum_{j \in S} |c_j|^2 E_{\alpha,j}. \end{aligned} \quad (48)$$

The reverse inequality with constant one is automatic. If $\varepsilon \mathfrak{L}_ \geq 1$, the displayed lower bound is interpreted as the trivial nonnegative bound.*

Proof. For each (α, j) , the deleted energy is at most

$$\#\{\gamma : M(\alpha, \gamma) = 0\} \max_{\gamma} w_\gamma \|f_{\alpha,j}(\gamma)\|^2 \leq \varepsilon \mathfrak{L}_{\alpha,j} E_{\alpha,j}. \quad (49)$$

Subtract this estimate, multiply by $|c_j|^2$, and sum in j . Deleting nonnegative terms proves the upper bound. $\square$

Remark 3.7 (Scope of the leverage condition). The exact lift (42) is purely Hilbert-space algebra. A bound $\mathfrak{L}_* = O(1)$ is a separate sampling assertion. Later sections verify an aggregate version on uniformly active prime-squarefree row blocks. Neither the identity nor the leverage estimate is a statement about coherent prime summation, sieve parity, or a fixed-shift prime-pair lower bound.

4 The exact diagonal Mellin ledger

We now evaluate the canonical envelope on the distinguished physical direction. The result is an exact coordinatewise identity. It uses neither orthogonality between different Mellin parameters nor a lower frame for arbitrary signed coefficient densities.

4.1 One direct cell and q smooth factors

Fix an integer $q \geq 1$, and let $0 \neq W_\nu \in C_c^\infty((0, \infty))$ for $1 \leq \nu \leq q$. We use the real-axis Mellin convention

$$\widetilde{W}_\nu(t) := \int_0^\infty W_\nu(x) x^{-it} \frac{dx}{x}, \quad (50)$$

$$W_\nu(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widetilde{W}_\nu(t) x^{it} dt. \quad (51)$$

Indeed, $s \mapsto W_\nu(e^s)$ is a compactly supported smooth function on $\mathbb{R}$, so $\widetilde{W}_\nu$ is Schwartz and (51) is ordinary Fourier inversion in the logarithmic coordinate.

Let $w_B : \mathbb{R}^q \rightarrow [1, \infty)$ be measurable and of at most polynomial growth; more precisely, for a fixed $B \geq 0$, assume

$$w_B(\mathbf{t}) \leq C_B(1 + |\mathbf{t}|)^B. \quad (52)$$

Define the positive product measure

$$d\mu_{\mathbf{W}}(\mathbf{t}) := (2\pi)^{-q} \prod_{\nu=1}^q |\widetilde{W}_\nu(t_\nu)| dt \quad (53)$$

and the polar phase

$$\varsigma_{\mathbf{W}}(\mathbf{t}) := \prod_{\nu=1}^q \begin{cases} \widetilde{W}_\nu(t_\nu)/|\widetilde{W}_\nu(t_\nu)|, & \widetilde{W}_\nu(t_\nu) \neq 0, \\ 0, & \widetilde{W}_\nu(t_\nu) = 0. \end{cases} \quad (54)$$

Its value on the $\mu_{\mathbf{W}}$ -null zero set is immaterial. The fixed weighted Wiener–Mellin mass is

$$A_B(\mathbf{W}) := \int_{\mathbb{R}^q} w_B(\mathbf{t}) d\mu_{\mathbf{W}}(\mathbf{t}) = \frac{1}{(2\pi)^q} \int_{\mathbb{R}^q} w_B(\mathbf{t}) \prod_{\nu=1}^q |\widetilde{W}_\nu(t_\nu)| dt. \quad (55)$$

Schwartz decay and (52) give

$$0 < A_B(\mathbf{W}) < \infty. \quad (56)$$

Let $\mathcal{I}$ be a finite nonempty coordinate set. For $i \in \mathcal{I}$, fix $c_i \in \mathbb{C}$ and positive samples $x_{i,1}, \dots, x_{i,q} > 0$. Put

$$A_{\mathbf{t}}(c)_i := \varsigma_{\mathbf{W}}(\mathbf{t}) c_i \prod_{\nu=1}^q x_{i,\nu}^{it_\nu} \in \ell^2(\mathcal{I}). \quad (57)$$

Every sample phase in (57) has unit modulus.

Theorem 4.1 (Exact q -factor diagonal Mellin identity). *For $c = (c_i)_{i \in \mathcal{I}} \in \ell^2(\mathcal{I})$, define*

$$\mathcal{S}_{\mathbf{W}} c := \int_{\mathbb{R}^q} A_{\mathbf{t}}(c) d\mu_{\mathbf{W}}(\mathbf{t}), \quad (58)$$

$$\mathcal{E}_B(\mathbf{W}; c) := \int_{\mathbb{R}^q} w_B(\mathbf{t}) \|A_{\mathbf{t}}(c)\|_{\ell^2(\mathcal{I})} d\mu_{\mathbf{W}}(\mathbf{t}). \quad (59)$$

Then the Bochner integral is absolutely convergent and

$$(\mathcal{S}_{\mathbf{W}} c)_i = c_i \prod_{\nu=1}^q W_\nu(x_{i,\nu}), \quad (60)$$

$$\|\mathcal{S}_{\mathbf{W}} c\|_2^2 = \sum_{i \in \mathcal{I}} |c_i|^2 \prod_{\nu=1}^q |W_\nu(x_{i,\nu})|^2, \quad (61)$$

$$\mathcal{E}_B(\mathbf{W}; c) = A_B(\mathbf{W}) \|c\|_2. \quad (62)$$

For $c \neq 0$, define the physical profile-overlap ratio

$$\Theta(\mathbf{W}; c) := \frac{\sum_{i \in \mathcal{I}} |c_i|^2 \prod_{\nu=1}^q |W_\nu(x_{i,\nu})|^2}{\sum_{i \in \mathcal{I}} |c_i|^2}. \quad (63)$$

If $\Theta(\mathbf{W}; c) > 0$, then

$$\boxed{\frac{\mathcal{E}_B(\mathbf{W}; c)^2}{\|\mathcal{S}_{\mathbf{W}} c\|_2^2} = \frac{A_B(\mathbf{W})^2}{\Theta(\mathbf{W}; c)}}. \quad (64)$$

If $c \neq 0$ and $\Theta(\mathbf{W}; c) = 0$, then $\mathcal{S}_{\mathbf{W}} c = 0$ while $\mathcal{E}_B(\mathbf{W}; c) > 0$; the ratio in (64) is therefore $+\infty$, in agreement with the extended-real convention $A_B^2/0 = +\infty$. For $c = 0$, both sides of the synthesis and envelope identities are zero, but no condition number and no ratio $0/0$ is defined.

Proof. Since $\mathcal{I}$ is finite, (56) and

$$\|A_{\mathbf{t}}(c)\|_2 = \|c\|_2 \quad \text{for } \mu_{\mathbf{W}}\text{-almost every } \mathbf{t} \quad (65)$$

give Bochner integrability. In coordinate i , Fubini's theorem and the product form of $\mu_{\mathbf{W}}$ give

$$\begin{aligned} \int_{\mathbb{R}^q} A_{\mathbf{t}}(c)_i d\mu_{\mathbf{W}}(\mathbf{t}) &= \frac{c_i}{(2\pi)^q} \int_{\mathbb{R}^q} \prod_{\nu=1}^q \widetilde{W}_\nu(t_\nu) x_{i,\nu}^{it_\nu} d\mathbf{t} \\ &= c_i \prod_{\nu=1}^q W_\nu(x_{i,\nu}), \end{aligned} \quad (66)$$

where the last line is (51) in each variable. This proves (60); orthogonality of the coordinate basis proves (61). Equation (65) and the definition (55) give (62). Finally, (63) says exactly that

$$\|\mathcal{S}_{\mathbf{W}} c\|_2^2 = \Theta(\mathbf{W}; c) \|c\|_2^2. \quad (67)$$

Combining this with (62) proves the ledger and all zero-denominator assertions. $\square$

Remark 4.2 (What the identity does not say). The coefficient vector c in [theorem 4.1](#) is the fixed base vector carried by the physical coordinates. The theorem does not permit an arbitrary signed density in $\mathbf{t}$, and it does not assert a lower frame on the full Mellin coefficient space. All row, orbit, and residue phases that have unit modulus may be included in the samples or in c_i ; literal mask zeros may also be included in c_i . The identity fails to describe a later map that first sums several physical coordinates coherently and only then takes a Hilbert norm.

4.2 A fixed number of direct residue cells

The actual provenance uses a fixed finite disjoint union of direct residue cells. Their output supports are orthogonal because the orbit classes are disjoint. The outer canonical envelope, however, is an ℓ^1 -sum over the cells. The following comparison records the resulting fixed constant exactly.

Corollary 4.3 (Finite direct-cell L^1 comparison). *Let $\mathfrak{A}$ be a fixed nonempty finite set. For every $a \in \mathfrak{A}$, let $\mathbf{W}_a = (W_{a,1}, \dots, W_{a,q_a})$ satisfy the hypotheses of [theorem 4.1](#), let $c^{(a)}$ be a finite base vector, and suppose that the corresponding physical outputs lie in pairwise orthogonal Hilbert subspaces. Put*

$$b_a := \|c^{(a)}\|_2, \quad A_a := A_B(\mathbf{W}_a), \quad (68)$$

$$\mathcal{E}_B^{\text{dir}} := \sum_{a \in \mathfrak{A}} A_a b_a, \quad \mathcal{B}_{\text{dir}}^2 := \sum_{a \in \mathfrak{A}} b_a^2, \quad (69)$$

$$\Theta_{\text{dir}} := \frac{\sum_{a \in \mathfrak{A}} \|\mathcal{S}_{\mathbf{W}_a} c^{(a)}\|_2^2}{\mathcal{B}_{\text{dir}}^2}, \quad (70)$$

whenever $\mathcal{B}_{\text{dir}} > 0$. Then

$$A_-^2 \mathcal{B}_{\text{dir}}^2 \leq (\mathcal{E}_B^{\text{dir}})^2 \leq A_2^2 \mathcal{B}_{\text{dir}}^2, \quad A_- := \min_{a \in \mathfrak{A}} A_a, \quad A_2^2 := \sum_{a \in \mathfrak{A}} A_a^2. \quad (71)$$

If $\Theta_{\text{dir}} > 0$, orthogonality of the cell outputs gives

$$\boxed{\frac{A_-^2}{\Theta_{\text{dir}}} \leq \frac{(\mathcal{E}_B^{\text{dir}})^2}{\sum_{a \in \mathfrak{A}} \|\mathcal{S}_{\mathbf{W}_a} c^{(a)}\|_2^2} \leq \frac{A_2^2}{\Theta_{\text{dir}}}}. \quad (72)$$

If $\Theta_{\text{dir}} = 0$, the nonzero base vector has zero physical output and positive envelope, so the middle ratio is $+\infty$. The constants A_- and A_2 depend only on the fixed direct cells, shapes, and profile weight, and not on X or on the physical coordinate multiplicities.

Proof. Every A_a is positive by (56). Therefore

$$\mathcal{E}_B^{\text{dir}} = \sum_a A_a b_a \geq A_- \sum_a b_a \geq A_- \left(\sum_a b_a^2 \right)^{1/2}. \quad (73)$$

Cauchy–Schwarz gives

$$\mathcal{E}_B^{\text{dir}} \leq \left(\sum_a A_a^2 \right)^{1/2} \left(\sum_a b_a^2 \right)^{1/2}. \quad (74)$$

These are (71). Pairwise orthogonality and (70) give

$$\sum_a \|\mathcal{S}_{\mathbf{W}_a} c^{(a)}\|_2^2 = \Theta_{\text{dir}} \mathcal{B}_{\text{dir}}^2. \quad (75)$$

Divide (71) by this identity. If $\Theta_{\text{dir}} = 0$, every physical cell output is zero while $\mathcal{E}_B^{\text{dir}} > 0$, proving the last assertion. $\square$

Remark 4.4 (Direct cells are essential). The orthogonality in [corollary 4.3](#) holds for direct residue classes, such as disjoint classes of $j \pmod{H_0}$. It is not asserted for overlapping signed Fourier cells or for projective-mask parameters. Appending cancelling signed layers can change their outer L^1 envelope without changing the physical multiplier. Thus the finite-cell comparison does not license a reverse Minkowski inequality for an arbitrary signed representation.

4.3 Identification with the retained physical coordinates

For the interface of [section 2](#), take

$$i = (m, r, \gamma, j), \quad (76)$$

include the literal mask, source coefficient, fixed residue amplitude, and common carrier in c_i , and take the positive samples $x_{i,\nu}$ to be the declared arguments of the physical smooth factors. Typical examples are $m_\gamma j/X$ and $d_\gamma j/K$. Before coherent prime summation and before grouping the residual label, these coordinates are orthogonal by (13). Hence [theorem 4.1](#) and [corollary 4.3](#) apply literally to the provenance-fixed direction $\mathcal{S}_X^L \mathbf{1} = G_X^L$.

The only remaining analytic input at this interface is a lower bound for the explicit nonnegative overlap Θ_{dir} . The diagonal theorem neither assumes nor proves such a bound. In particular, it contains no distribution estimate for primes, no parity-sensitive cancellation, and no fixed-shift prime-pair conclusion.

5 Uniform profile overlap on sampled orbit fibers

The diagonal Mellin identity reduces the actual-direction problem to a nonnegative comparison: the sampled energy of the unseparated physical profile must not be much smaller than the energy of the unimodular Mellin carrier. This section proves that comparison under one explicit geometric hypothesis. It is important that the hypothesis is stated separately from the row-flatness theorem inherited from TPC–50.

5.1 A row–orbit sampling interface

Put $T = \log 2$ and $Y = [0, T]^2$. Let $\mathcal{R}_X$ be a finite opposite-row set, with logarithmic coordinates

$$q_\gamma = (x_\gamma, y_\gamma) \in Y \quad (\gamma \in \mathcal{R}_X),$$

positive weights $\omega_{\gamma, X}$, and total weight

$$E_X := \sum_{\gamma \in \mathcal{R}_X} \omega_{\gamma, X} > 0. \quad (77)$$

We use the following three hypotheses.

(S1) The probability measures

$$\nu_X := E_X^{-1} \sum_{\gamma \in \mathcal{R}_X} \omega_{\gamma, X} \delta_{q_\gamma} \quad (78)$$

converge weakly to a probability measure ν on Y . In the prime–squarefree application one takes the exact base weight $\omega_{\gamma, X} = (\log \ell_\gamma)^{2e}$, $e \in \{0, 1\}$, and places every fixed dyadic or unary shape in B and F . Then $d\nu = e^{x+y} dx dy$ in the limit (this density already has mass one on Y) [6]. With any other base weight, both weak convergence and the identity of its limiting measure are separate hypotheses.

(S2) The base weights are uniformly flat:

$$|\mathcal{R}_X| \max_{\gamma \in \mathcal{R}_X} \omega_{\gamma, X} \leq C_{\text{flat}} E_X. \quad (79)$$

(S3) The literal mask $M_X(\alpha, \gamma) \in \{0, 1\}$ obeys the one-sided degree estimate

$$\sup_{\alpha} \#\{\gamma \in \mathcal{R}_X : M_X(\alpha, \gamma) = 0\} \leq \epsilon_X |\mathcal{R}_X|, \quad \epsilon_X = o(1). \quad (80)$$

For the TPC–50/51 application, $\mathcal{R}_X$ is required to be one complete positive-density admissible prime–squarefree row cell after fixed residue subdivision, in the nondegenerate interior regime used there (in particular the inherited size separation and injectivity of $m = \ell d$ hold). No independent pruning of the opposite row is allowed before (S3) is applied. If a declared packet does not have this form, (S1)–(S3) are new assumptions rather than consequences of TPC–50/51.

The source label α is not assigned a probability measure. It may be supported on one row, and later it may carry an arbitrary cofactor multiplicity. This asymmetry is what permits the final theorem to be uniform under source and r -concentration.

Let $\mathfrak{B}$ be a compact parameter space containing all normalized source-scale parameters. Let $\mathcal{A}_0$ be a fixed finite collection of direct orbit-residue cells. To each $a \in \mathcal{A}_0$ and $\beta \in \mathfrak{B}$ associate a carrier and a physical profile

$$B_{\beta, a}, F_{\beta, a} \in C^1(Y \times I_0). \quad (81)$$

We assume that both families are compact in $C^1(Y \times I_0)$. In particular, their sup norms and first derivatives are uniformly bounded. The direct cell a corresponds to the progression $j \equiv a \pmod{H_0}$; cells with zero physical residue amplitude are removed before this definition.

Define the continuous carrier and profile energies by

$$\mathcal{B}(\beta) := \frac{1}{H_0} \sum_{a \in \mathcal{A}_0} \int_Y \int_{I_0} |B_{\beta,a}(q, t)|^2 dt d\nu(q), \quad (82)$$

$$\mathcal{P}(\beta) := \frac{1}{H_0} \sum_{a \in \mathcal{A}_0} \int_Y \int_{I_0} |F_{\beta,a}(q, t)|^2 dt d\nu(q). \quad (83)$$

We impose the carrier nondegeneracy and normalization

$$0 < b_- \leq \mathcal{B}(\beta) \leq b_+ < \infty \quad (\beta \in \mathfrak{B}). \quad (84)$$

Deleting an identically zero direct cell is harmless, but the uniform lower bound $b_- > 0$ is a genuine compact-packet hypothesis; it is not supplied by row flatness alone. Once that lower bound is known, rescaling fixes the upper and lower constants.

Assumption 5.1 (Uniform physical profile nondegeneracy). *The declared direct packet satisfies*

$$\boxed{\rho_* := \inf_{\beta \in \mathfrak{B}} \frac{\mathcal{P}(\beta)}{\mathcal{B}(\beta)} > 0.} \quad (85)$$

Assumption 5.1 is new. It is not a consequence of TPC-50. That paper proves flatness and literal-mask survival for an identified fixed-complexity unary row factor, uniformly in one real-axis Mellin phase. It does not prove that the complete product of source, opposite-row, and orbit cutoffs retains a positive amount of carrier energy after the Mellin variables are reassembled. In particular, it does not exclude two scaled compact supports approaching a common flat endpoint. For named fixed cutoffs, (85) is a deterministic support-geometry condition on a compact finite-dimensional parameter set; it is not a prime-pair hypothesis.

Lemma 5.2 (A nonempty active subcell from one strict overlap). *Assume $\mathfrak{B} \subset \mathbb{R}^d$, the maps $\beta \mapsto B_{\beta,a}, F_{\beta,a}$ are continuous in $C(Y \times I_0)$, and (84) holds. If, for some interior parameter β_0 , some direct cell a_0 , and some $(q_0, t_0) \in \text{supp } \nu \times \text{int } I_0$,*

$$|F_{\beta_0,a_0}(q_0, t_0)| > 0, \quad (86)$$

then there is a nonempty compact parameter cell $\mathfrak{B}_0 \Subset \mathfrak{B}$, containing β_0 in its relative interior, on which

$$\inf_{\beta \in \mathfrak{B}_0} \frac{\mathcal{P}(\beta)}{\mathcal{B}(\beta)} > 0. \quad (87)$$

Proof. Continuity and (86) give an open row-orbit box of positive $\nu \otimes dt$ measure on which the same profile remains nonzero. Hence $\mathcal{P}(\beta_0) > 0$. The two energy functions are continuous in β , while $\mathcal{B} \geq b_- > 0$. Their ratio is therefore continuous and positive on a sufficiently small neighborhood of β_0 . Choose a compact cell inside that neighborhood. $\square$

The lemma proves existence only after a strict overlap point has been verified for the named physical cutoffs. It does not claim that every dyadic boundary packet, or the closure of the full inherited parameter range, is uniformly active.

For $\beta \in \mathfrak{B}$, define the unmasked sampled energies

$$\mathcal{B}_X^0(\beta) := \frac{1}{E_X J} \sum_{a \in \mathcal{A}_0} \sum_{\gamma \in \mathcal{R}_X} \omega_{\gamma,X} \sum_{j \in \mathcal{J}_{X,a}} |B_{\beta,a}(q_\gamma, j/J)|^2, \quad (88)$$

$$\mathcal{P}_X^0(\beta) := \frac{1}{E_X J} \sum_{a \in \mathcal{A}_0} \sum_{\gamma \in \mathcal{R}_X} \omega_{\gamma,X} \sum_{j \in \mathcal{J}_{X,a}} |F_{\beta,a}(q_\gamma, j/J)|^2. \quad (89)$$

For a source row α , carrying a parameter $\beta_\alpha \in \mathfrak{B}$, insert $M_X(\alpha, \gamma)$ in these sums and denote the resulting quantities by $\mathcal{B}_X^M(\alpha)$ and $\mathcal{P}_X^M(\alpha)$.

Lemma 5.3 (Uniform row-orbit sampling). *Under (S1) and the compact C^1 profile hypotheses,*

$$\sup_{\beta \in \mathfrak{B}} |\mathcal{B}_X^0(\beta) - \mathcal{B}(\beta)| = o(1), \quad (90)$$

$$\sup_{\beta \in \mathfrak{B}} |\mathcal{P}_X^0(\beta) - \mathcal{P}(\beta)| = o(1). \quad (91)$$

Proof. For a uniformly bounded subset of $C^1(I_0)$, progression Riemann sums obey

$$\frac{1}{J} \sum_{j \in \mathcal{J}_{X,a}} h(j/J) = \frac{1}{H_0} \int_{I_0} h(t) dt + O(J^{-1})$$

uniformly in h . Apply this with $h(t) = |B_{\beta,a}(q, t)|^2$ and with F in place of B . The resulting functions of q , as β and a vary, form compact subsets of $C(Y)$.

Weak convergence of probability measures is uniform on a compact subset of $C(Y)$: given $\delta > 0$, choose a finite δ -net in the sup norm, use weak convergence on its finitely many elements, and then use that both measures have mass one. Applying this observation to the preceding compact families, and then summing over the fixed set $\mathcal{A}_0$, proves both claims. $\square$

Lemma 5.4 (Literal-mask deletion). *Under (S2)–(S3), uniformly in the source row α ,*

$$0 \leq \mathcal{B}_X^0(\beta_\alpha) - \mathcal{B}_X^M(\alpha) \leq C\epsilon_X, \quad (92)$$

$$0 \leq \mathcal{P}_X^0(\beta_\alpha) - \mathcal{P}_X^M(\alpha) \leq C\epsilon_X, \quad (93)$$

where C depends only on the fixed profile families, residue cells, and the flatness constant.

Proof. For each deleted opposite row, the orbit sum of either squared profile is $O(J)$, uniformly in α, γ , by compactness in C^1 . There are at most $\epsilon_X |\mathcal{R}_X|$ deleted rows. Hence the unnormalized loss is at most

$$CJ\epsilon_X |\mathcal{R}_X| \max_{\gamma} \omega_{\gamma,X} \leq CC_{\text{flat}} \epsilon_X J E_X$$

by (79). Division by $J E_X$ proves the two estimates. $\square$

Theorem 5.5 (Uniform sampled overlap under the literal mask). *Assume (S1)–(S3), (84), and Assumption 5.1. Then, for all sufficiently large X , uniformly in every source row α ,*

$$\mathcal{B}_X^M(\alpha) \geq \frac{b_-}{2} > 0, \quad \frac{\mathcal{P}_X^M(\alpha)}{\mathcal{B}_X^M(\alpha)} \geq \frac{\rho_*}{2}. \quad (94)$$

Proof. By lemmas 5.3 and 5.4, both masked sampled energies differ uniformly from their continuous counterparts by $o(1)$. In particular, the carrier is at least $b_-/2$ for large X . Also $\mathcal{P}(\beta) \geq \rho_* \mathcal{B}(\beta) \geq \rho_* b_-$. Choose X so large that the two absolute errors are smaller than a fixed sufficiently small multiple of $\rho_* b_-$. Dividing the resulting profile lower bound by the corresponding carrier upper bound gives the stated constant $\rho_*/2$, after harmlessly enlarging the threshold for X . $\square$

The number $1/2$ is inessential. The proof gives $\rho_* + o(1)$ after the carrier normalization is kept in the ratio. The weaker displayed constant is convenient for later direct sums.

5.2 Arbitrary outer concentration

Let $\mathcal{U}_X$ be any finite source/cofactor label set. Each $u \in \mathcal{U}_X$ carries a source row $\alpha(u)$, a parameter $\beta_{\alpha(u)}$, and an arbitrary scalar or Hilbert coefficient a_u . Define

$$\mathcal{C}_X(a) := E_X J \sum_{u \in \mathcal{U}_X} \|a_u\|^2 \mathcal{B}_X^M(\alpha(u)), \quad (95)$$

$$\mathcal{P}_X(a) := E_X J \sum_{u \in \mathcal{U}_X} \|a_u\|^2 \mathcal{P}_X^M(\alpha(u)). \quad (96)$$

Corollary 5.6 (Source- and cofactor-uniform overlap). *Under the hypotheses of [theorem 5.5](#),*

$$\boxed{\mathcal{P}_X(a) \geq \frac{\rho_*}{2} \mathcal{C}_X(a)} \quad (97)$$

for every finitely supported family (a_u) . No lower source count, source equidistribution, or uniform cofactor multiplicity is required.

Proof. Apply (94) separately to every retained source/cofactor fiber, multiply by $\|a_u\|^2$, and sum. All terms are nonnegative because the source and cofactor labels have not been coherently grouped. $\square$

5.3 Good and bad overlap cells

Uniform positivity may fail on the closure of a packet even though most carrier energy lies away from the grazing boundary. The following threshold form records exactly what an energy selection must prove.

For $\eta > 0$, put

$$\mathfrak{B}_{\text{good}}(\eta) := \left\{ \beta \in \mathfrak{B} : \frac{\mathcal{P}(\beta)}{\mathcal{B}(\beta)} \geq \eta \right\}, \quad \mathfrak{B}_{\text{bad}}(\eta) := \mathfrak{B} \setminus \mathfrak{B}_{\text{good}}(\eta). \quad (98)$$

Let Δ_X^{samp} denote the maximum of the two uniform sampling errors in [lemma 5.3](#) plus $C\epsilon_X$, with the constant from [lemma 5.4](#).

Proposition 5.7 (Threshold overlap tradeoff). *Let $\eta_X > 0$ satisfy*

$$\Delta_X^{\text{samp}} = o(\eta_X). \quad (99)$$

For all sufficiently large X , every source fiber whose parameter lies in $\mathfrak{B}_{\text{good}}(\eta_X)$ has sampled masked profile-to-carrier ratio at least $\eta_X/2$.

For an arbitrary outer family a , let $\mathcal{C}_X^{\text{good}}(a)$ and $\mathcal{C}_X^{\text{bad}}(a)$ be the carrier energies from (95) restricted according to (98), and put

$$\sigma_X := \frac{\mathcal{C}_X^{\text{good}}(a)}{\mathcal{C}_X(a)} \quad \text{when } \mathcal{C}_X(a) > 0. \quad (100)$$

Then

$$\boxed{\mathcal{P}_X(a) \geq \frac{\eta_X \sigma_X}{2} \mathcal{C}_X(a).} \quad (101)$$

Proof. The proof of [theorem 5.5](#) is uniform on the good set. Condition (99) makes its absolute sampling-and-deletion error negligible relative to $\eta_X \mathcal{B}(\beta)$, since the carrier is at least b_- . This gives the fiberwise factor $\eta_X/2$. Sum only the good fibers as in [corollary 5.6](#); the bad contribution to the physical energy is nonnegative and may be discarded. The definition of σ_X then gives (101). $\square$

Mere weak convergence in (S1) proves the proposition for a fixed $\eta > 0$, but it does not by itself verify (99) for a polynomially shrinking η_X . Such a use requires a quantitative row-sampling theorem at the resolution of the shrinking overlap, together with a mask error smaller than η_X . This distinction prevents a qualitative compactness argument from being misreported as a polynomial endpoint estimate.

For clarity, the quantitative fixed-modulus sampling error available in the inherited row theorem contains $e^{-c\sqrt{\log L}}$. This tends to zero but is larger than $X^{-\lambda}$ for every fixed $\lambda > 0$. Hence it does not verify $\Delta_X^{\text{samp}} = o(X^{-\lambda})$ for a polynomially shrinking threshold. A strict inequality $\lambda < 1/400$ controls the inherited mask-deletion term only; the shrinking-threshold branch still requires a sharper relative sampling theorem (or a fixed positive overlap).

5.4 The exponent ledger

Suppose a displayed Mellin representation satisfies

$$\mathcal{E}_X^2 \leq X^{\omega+o(1)} \mathcal{C}_X(a), \quad (102)$$

and write

$$\eta_X = X^{-\lambda+o(1)}, \quad \sigma_X = X^{-\varsigma+o(1)}. \quad (103)$$

Then [proposition 5.7](#) gives

$$\frac{\mathcal{E}_X^2}{\mathcal{P}_X(a)} \leq X^{\omega+\lambda+\varsigma+o(1)}. \quad (104)$$

For a family of $R_X = X^{\varrho+o(1)}$ direct cells whose individual weighted Mellin masses are at most $X^{\tau+o(1)}$, Cauchy–Schwarz in the finite cell index gives the admissible ledger

$$\omega \leq 2\tau + \varrho. \quad (105)$$

Thus a sufficient endpoint condition is

$$\boxed{2\tau + \varrho + \lambda + \varsigma \leq \frac{1}{400}}, \quad (106)$$

in addition to the relative sampling requirement [\(99\)](#). For a declared fixed collection of direct residue cells, fixed smooth shapes, and fixed derivative order, $\tau = \varrho = 0$. Under the additional exact-profile identification and uniform nondegeneracy hypotheses used below, $\lambda = \varsigma = 0$ as well, so this gate consumes no positive part of the $1/400$ allowance.

At the inherited endpoint the degree defect contains a term $X^{-1/400+o(1)}$. Consequently a threshold exactly of size $X^{-1/400+o(1)}$ is not automatically larger than the mask error. A strict exponent margin, or a sharper packet-specific deletion theorem, is needed before equality at that boundary may be claimed.

6 The coefficientwise actual-direction bridge

We now combine the sampled overlap theorem with the provenance-fixed Mellin representation. The conclusion concerns one distinguished coefficient density and the coefficientwise pre-terminal Hilbert space. It is not a lower-frame theorem for arbitrary signed Mellin densities.

6.1 The exact actual-to-profile dictionary

The sampled theorem of [section 5](#) is abstract until its weights and profiles are identified coordinate by coordinate with [\(13\)](#). We therefore isolate, rather than hide, the required application hypothesis. For notational economy it is stated on one fixed row-residue group; its finitely many orbit-residue cells form $\mathcal{A}_0$. A fixed finite collection of row-residue groups is treated by orthogonal summation.

Assumption 6.1 (Exact actual-profile identification). *The following data agree on every retained coordinate $i = (a, m, r, j, \gamma)$.*

- (I1) *The opposite-row universe is one complete positive-density admissible prime-squarefree residue cell in the nondegenerate interior regime of TPC-50, $m = \ell d$ is injective there, and no independent opposite-row pruning has occurred. The only deletion is $M_X(m, \gamma) = K_{m\gamma}$.*
- (I2) *The base weight is $\omega_{\gamma, X} = (\log \ell_\gamma)^{2e}$, $e \in \{0, 1\}$. Every fixed dyadic, unary, and residue amplitude not contained in the outer coefficient is placed in the named functions $B_{\beta_m, a}$ and $F_{\beta_m, a}$.*

(I3) *There are outer source coefficients a_m , independent of the retained cofactor label r , and diagonal carrier coordinates $c_{X,a}(i)$ such that*

$$|c_{X,a}(i)|^2 = M_X(m, \gamma) |a_m|^2 \omega_{\gamma,X} |B_{\beta_m,a}(q_\gamma, j/J)|^2, \quad (107)$$

$$|\gamma_m^{(1)}|^2 K_{m\gamma} |\Xi_{m,\gamma}(j)|^2 |\mathcal{W}_{m,\gamma}(j/J)|^2 = M_X(m, \gamma) |a_m|^2 \omega_{\gamma,X} |F_{\beta_m,a}(q_\gamma, j/J)|^2. \quad (108)$$

The direct cells are disjoint and these identities use the same physical atomic normalization as (10).

This assumption is a checkable dictionary, not a conclusion of the cited papers. TPC–25 supplies the constituent Mellin factorizations and TPC–50/51 supply the complete-row sampling and mask estimates on their declared cells; neither reference proves that an arbitrary inherited packet satisfies all three items above. If the dictionary is not verified, the results below remain conditional analytic criteria for that packet.

6.2 Carrier energy and the diagonal layer norm

Let $\mathcal{U}_X$ be the finite set of retained source/cofactor labels $u = (m, r)$ in the fixed row-residue group. For each $u = (m, r)$, put $a_u := a_m$, where a_m contains the literal source coefficient and every factor independent of the opposite row and orbit coordinates. Thus the same source amplitude is repeated on every retained cofactor fiber, as in (11). The opposite-row base weight is $\omega_{\gamma,X}$, and the literal mask is $M_X(m, \gamma)$. With the notation of section 5, define the unnormalized carrier energy

$$\mathcal{C}_X^L := E_X J \sum_{u=(m,r) \in \mathcal{U}_X} |a_u|^2 \mathcal{B}_X^M(m). \quad (109)$$

The physical profile energy is

$$\mathcal{P}_X^L := E_X J \sum_{u=(m,r) \in \mathcal{U}_X} |a_u|^2 \mathcal{P}_X^M(m). \quad (110)$$

Here the same source row may occur with any number of cofactors. No uniformity of that multiplicity is assumed.

Whenever $\mathcal{C}_X^L > 0$, define the aggregate actual overlap

$$\Theta_X^{\text{act}} := \frac{\mathcal{P}_X^L}{\mathcal{C}_X^L}. \quad (111)$$

On a single direct cell this is the sampled instantiation of $\Theta(\mathbf{W}; c)$ from (63); for a fixed finite cell family it is the corresponding carrier-energy-weighted aggregate.

Summing (108) over the disjoint physical coordinates, and using $K_{m\gamma}^2 = K_{m\gamma}$, gives

$$\boxed{\mathcal{P}_X^L = \|G_X^L\|_{\mathcal{H}_X^{\text{coeff}}}^2 = \|\mathcal{S}_X^L \mathbf{1}\|_{\mathcal{H}_X^{\text{coeff}}}^2.} \quad (112)$$

No source sum, prime sum, or residual grouping occurs in (112). In particular, this equality is licensed by the displayed coordinate dictionary, not by a change of normalization after the fact.

The point of real-axis Mellin inversion is that its scale factors have unit modulus. More precisely, after the absolute values of the fixed Mellin transform densities have been placed in $d\mu_X$, every continuous layer on a direct cell has the form

$$A_{X,\theta}^L(i) = c_{X,a}(i) \exp(i\Phi_{X,a,\theta}(i)), \quad |\exp(i\Phi_{X,a,\theta}(i))| = 1, \quad (113)$$

on its finite coordinate set. The vector $c_{X,a}$ includes the literal mask and the common compact orbit carrier, but not the physical factors that are reconstructed by Mellin inversion. Therefore

$$\|A_{X,\theta}^L\| = R_{X,a} \quad (114)$$

is independent of the continuous Mellin parameter on that cell, and

$$\mathcal{C}_X^L = \sum_{a \in \mathcal{A}_0} R_{X,a}^2. \quad (115)$$

For each nonzero cell put

$$M_{a,B} := \int_{\Theta_a} w_B(\theta) d\mu_{X,a}(\theta). \quad (116)$$

The scale parameters occur only in the phases in (113). The transform densities come from a fixed finite library of nonzero smooth shapes; more generally their weighted total-variation masses depend continuously on normalized parameters in a fixed compact set. Schwartz decay gives the upper bound below, while nonzero mass and compactness give the lower bound. Thus there are constants

$$0 < m_B \leq M_{a,B} \leq M_B < \infty \quad (117)$$

for all active cells and every X . A zero-measure template is discarded before the active-cell family is declared.

Proposition 6.2 (Finite-cell carrier identity). *For the provenance-fixed direction,*

$$\mathcal{E}_{1,B}^L(\mathbf{1}) = \sum_{a \in \mathcal{A}_0} M_{a,B} R_{X,a}. \quad (118)$$

Consequently there are fixed constants $0 < c_B \leq C_B < \infty$ such that

$$c_B \mathcal{C}_X^L \leq (\mathcal{E}_{1,B}^L(\mathbf{1}))^2 \leq C_B \mathcal{C}_X^L. \quad (119)$$

Proof. Equation (114) makes the norm in the definition of $\mathcal{E}_{1,B}^L$ independent of the continuous Mellin parameter. Integrate on each direct cell and then sum over the finite disjoint union to obtain (118). The lower bound follows from

$$\left(\sum_a M_{a,B} R_{X,a} \right)^2 \geq m_B^2 \sum_a R_{X,a}^2.$$

For the upper bound, Cauchy–Schwarz in the fixed cell index gives

$$\left(\sum_a M_{a,B} R_{X,a} \right)^2 \leq \left(\sum_a M_{a,B}^2 \right) \left(\sum_a R_{X,a}^2 \right).$$

Use (117) and (115). □

This proposition identifies exactly what Schwartz decay, finite physical support, and Bochner integrability accomplish. They make the weighted Mellin mass finite and compare the outer $L^1(d\mu_X)$ envelope with the *layer carrier energy*. They do not compare that carrier with the signed reassembled physical output. The latter comparison is precisely the profile overlap proved in the preceding section.

6.3 The uniformly active actual direction

Theorem 6.3 (Conditional coefficientwise actual Mellin bridge). *Assume the inherited coefficient interface of section 2, Assumption 6.1, the row-sampling and literal-mask hypotheses (S1)–(S3), the finite direct-cell diagonal form (113), and Assumption 5.1, and suppose $\mathcal{C}_X^L > 0$. Then for all sufficiently large X , $\Theta_X^{\text{act}} \geq \rho_*/2$,*

$$\|S_X^L \mathbf{1}\|_{\mathcal{H}_X^{\text{coeff}}}^2 \geq \frac{\rho_*}{2} \mathcal{C}_X^L, \quad (120)$$

and

$$\kappa_{\text{can},B}^L(X) = \frac{(\mathcal{E}_{1,B}^L(\mathbf{1}))^2}{\|\mathcal{S}_X^L \mathbf{1}\|_{\mathcal{H}_X^{\text{coeff}}}^2} \leq \frac{2C_B}{\rho_*} = O_{B,H_0,\mathcal{A}_0,\mathbf{w}}(1). \quad (121)$$

The conclusion is uniform in arbitrary concentration of the actual source coefficients and arbitrary finite multiplicity of the retained cofactors r . The symmetric right coefficient has the same conclusion; a fixed orthogonal left–right sum changes only the displayed constant.

Proof. Use (112) and apply [corollary 5.6](#) with the actual outer family $a_{(m,r)} := a_m$. This proves (120). The upper inequality in (119) then gives (121). The proof is fiberwise before the outer direct sum, so neither source concentration nor cofactor multiplicity introduces a counting loss. $\square$

The theorem controls the one density fixed by provenance. It does not contradict the finite-rank kernel theorem for arbitrary continuous Mellin densities, and it does not produce a uniform lower frame on the entire growing translate family. The orbit, source, opposite-row, and cofactor labels prevent the artificial scalar cross terms responsible for that false frame problem.

6.4 A good/bad actual bridge

The threshold theorem has an immediate actual-direction form. Let $\mathcal{C}_{X,\text{good}}^L$ be the part of (109) on source fibers in $\mathfrak{B}_{\text{good}}(\eta_X)$, and set

$$\sigma_X^L := \frac{\mathcal{C}_{X,\text{good}}^L}{\mathcal{C}_X^L} \quad (\mathcal{C}_X^L > 0). \quad (122)$$

Corollary 6.4 (Energy-selected actual bridge). *Retain the dictionary, finite-cell, sampling, mask, and profile hypotheses of [theorem 6.3](#). Assume also $\mathcal{C}_X^L > 0$, $\sigma_X^L > 0$, and $\Delta_X^{\text{samp}} = o(\eta_X)$. Then*

$$\|\mathcal{S}_X^L \mathbf{1}\|^2 \geq \frac{\eta_X \sigma_X^L}{2} \mathcal{C}_X^L \quad (123)$$

and

$$\kappa_{\text{can},B}^L(X) \leq \frac{2C_B}{\eta_X \sigma_X^L}. \quad (124)$$

In particular, if $\eta_X = X^{-\lambda+o(1)}$ and $\sigma_X^L = X^{-\varsigma+o(1)}$, then

$$\kappa_{\text{can},B}^L(X) \leq X^{\lambda+\varsigma+o(1)}. \quad (125)$$

Proof. Equation (123) is [proposition 5.7](#) applied to the actual coefficients. Combine it with (119); the exponent form is immediate. $\square$

The factor σ_X^L cannot be omitted. If the actual source and cofactor mass is allowed to concentrate entirely on grazing fibers, a good set of large row density says nothing about the distinguished actual direction. Conversely, no equidistribution of the source is needed once a positive good-carrier fraction is established by a separate energy selection.

6.5 Endpoint cost

At the inherited endpoint,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad \kappa_0 = \frac{1}{400}. \quad (126)$$

For a fixed collection of direct residue cells, fixed smooth shapes, and a fixed derivative order B , the weighted Mellin masses in (117) have zero polynomial cost. Orbit sampling contributes

$O(J^{-1}) = X^{-133/400+o(1)}$, and the inherited literal-mask defect tends to zero. Hence a fixed $\rho_* > 0$ gives (121) with zero positive polynomial loss and spends none of the $1/400$ allowance.

For a shrinking good threshold and a shrinking good-carrier fraction, the exact cost is instead

$$\lambda + \varsigma \tag{127}$$

in the fixed-cell situation. More generally one adds the transform and cell-count cost $2\tau + \varrho$ from (105). Therefore the sufficient ledger is exactly (106). This exponent ledger does not verify its own relative sampling premise: a polynomially shrinking threshold still requires $\Delta_X^{\text{samp}} = o(\eta_X)$. A strict margin $\lambda < 1/400$ makes only the inherited mask-deletion term negligible. The available fixed-modulus row-sampling error $e^{-c\sqrt{\log L}}$ is not $o(X^{-\lambda})$ for any fixed $\lambda > 0$. Thus the polynomial threshold branch remains conditional on a sharper relative sampling theorem (or it must be replaced by fixed uniform overlap).

6.6 Why the TPC-48 profile does not follow

The present result is deliberately placed before the terminal prime map. In TPC-48, a fixed projective layer contains

$$B_{\omega,m}(\alpha) = \sum_{p,r} A_{\omega}(p,r,m) e(-pr\alpha) v_{\omega,p,r,m}, \tag{128}$$

and its displayed envelope has the order

$$\mathfrak{S}_{\mathcal{R}}(\alpha) = \int C_{\text{LS}}(\phi_{\omega}, H_0)^{1/2} \left(\sum_m \|B_{\omega,m}(\alpha)\|^2 \right)^{1/2} d|\nu|(\omega). \tag{129}$$

This is not $\mathcal{E}_{1,B}^L(\mathbf{1})$. Four operations intervene:

- (D1) the prime variable is summed coherently at fixed (m, r) before the Hilbert norm is taken;
- (D2) the outer total-variation integral is taken after a nonlinear square root of the layer energy;
- (D3) ω contains projective-mask and terminal parameters that are not direct physical Mellin cells; and
- (D4) $C_{\text{LS}}^{1/2}$ is a profile seminorm weight, not the norm of a canonical Mellin coefficient.

The diagonal identity survives unitary coordinate phases, but it does not survive an arbitrary coherent map with a kernel. Thus theorem 6.3 cannot be moved through (128) by Minkowski in reverse.

After the present bridge, the remaining interfaces are, in dependency order:

- (O1) a lower or two-sided coherent-prime square at fixed (m, r) , stable under the actual Mellin carrier projection;
- (O2) control of the many-to-one map grouping $s = d(m)r$, including its fiber multiplicity and possible cancellation;
- (O3) localization of the completed frozen shift family back to the one physical intercept h_0 ; and
- (O4) identification of the residual low-sign Gram with the native affine physical Gram without discarding its cross terms.

None is proved by profile nondegeneracy. In particular, the present theorem contains no fixed-shift prime-pair lower bound, no parity-breaking estimate, and no twin-prime conclusion.

7 Hilbert-valued row-log compression without orbit collapse

The preceding diagonal argument needs only the distinguished physical direction. It is nevertheless useful to locate that direction inside the finite row-log banks of TPC-51. The point of this section is that smoothness gives a uniform finite approximation in the two *row-log* variables while the complete orbit and residual fiber is left untouched. In particular, none of the statements below replaces the growing orbit space by a scalar space of polylogarithmic rank.

7.1 Tensor cosine projection

Put $T = \log 2$. On $[0, T]$, let

$$\chi_0(x) := T^{-1/2}, \quad \chi_k(x) := (2/T)^{1/2} \cos(\pi kx/T) \quad (k \geq 1), \quad (130)$$

and set

$$\chi_{k_1, k_2}(x, y) := \chi_{k_1}(x) \chi_{k_2}(y). \quad (131)$$

This is an orthonormal basis of $L^2([0, T]^2)$. If $\mathcal{K}$ is a Hilbert space and $H : [0, T]^2 \rightarrow \mathcal{K}$ is Bochner measurable, define

$$\Pi_{K_F}^{\text{row}} H := \sum_{0 \leq k_1, k_2 \leq K_F} \left(\int_{[0, T]^2} H(x, y) \chi_{k_1, k_2}(x, y) dx dy \right) \chi_{k_1, k_2}. \quad (132)$$

The coefficient in parentheses is a vector in $\mathcal{K}$.

For later pointwise sampling, use the mixed regularity norm

$$\|H\|_{\mathcal{C}^{2,2}} := \sum_{0 \leq a, b \leq 2} \|\partial_x^a \partial_y^b H\|_{L^\infty([0, T]^2; \mathcal{K})}. \quad (133)$$

The mixed derivatives in this definition are deliberately stronger than the H^1 regularity needed for the continuum L^2 tail. They give a uniform bound at the arithmetic sampling nodes, including the boundary of the dyadic log box.

Theorem 7.1 (Hilbert-valued tensor cosine compression). *There is a constant $C_T < \infty$, independent of $\mathcal{K}$, such that for every $H \in C^{2,2}([0, T]^2; \mathcal{K})$ and $K_F \geq 1$,*

$$\|H - \Pi_{K_F}^{\text{row}} H\|_{L^2(e^{x+y} dx dy; \mathcal{K})}^2 \leq \frac{C_T}{(K_F + 1)^2} \|H\|_{\mathcal{C}^{2,2}}^2, \quad (134)$$

$$\|H - \Pi_{K_F}^{\text{row}} H\|_{L^\infty([0, T]^2; \mathcal{K})}^2 \leq \frac{C_T}{(K_F + 1)^2} \|H\|_{\mathcal{C}^{2,2}}^2. \quad (135)$$

The same estimates hold for a finite or countable family $(H_\alpha)_\alpha$ after summing both sides in α , whenever the right side is finite.

Proof. For the L^2 assertion, first use Lebesgue measure. The tensor cosines diagonalize the Neumann Laplacian. Every omitted index has $k_1 > K_F$ or $k_2 > K_F$, and hence eigenvalue at least $(\pi(K_F + 1)/T)^2$. Parseval, applied after scalar pairing with an orthonormal basis of $\mathcal{K}$, gives

$$\|H - \Pi_{K_F}^{\text{row}} H\|_{L^2}^2 \leq \frac{T^2}{\pi^2(K_F + 1)^2} (\|\partial_x H\|_{L^2}^2 + \|\partial_y H\|_{L^2}^2).$$

The density e^{x+y} is bounded above and below on the fixed box, so this proves (134).

For the pointwise assertion, integrate the cosine coefficients twice by parts in every nonzero coordinate. The sine boundary terms vanish at the first integration; the second integration

produces only first derivatives at the endpoints and integrals of second derivatives. Consequently the Hilbert-valued coefficient $\widehat{H}(k_1, k_2)$ satisfies

$$\|\widehat{H}(k_1, k_2)\|_{\mathcal{K}} \leq \frac{C_T \|H\|_{\mathcal{C}^{2,2}}}{(1+k_1)^2(1+k_2)^2}.$$

The sum of this majorant over the complement of $\{0, \dots, K_F\}^2$ is $O_T((K_F + 1)^{-1})$. Absolute convergence of the cosine series, followed by squaring, proves (135). Every step is stable under orthogonal summation in $\mathcal{K}$, which also proves the family version. $\square$

7.2 Arithmetic sampling and literal-mask deletion

Let $\mathcal{R}_X$ be the complete prime-squarefree row universe and let $\mathcal{R}_X(\mathcal{A}) \subset \mathcal{R}_X$ be one fixed positive-density admissible row cell, with logarithmic nodes

$$z_\gamma = (x_\gamma, y_\gamma) = (\log(\ell_\gamma/L), \log(d_\gamma/D)),$$

base weights $w_\gamma = (\log \ell_\gamma)^{2e}$, where $e \in \{0, 1\}$, and total base energy $E_X = \sum_{\gamma \in \mathcal{R}_X(\mathcal{A})} w_\gamma$. We retain the inherited base flatness

$$E_X \asymp_{\mathcal{A}} |\mathcal{R}_X(\mathcal{A})| (\log L)^{2e}, \quad \max_{\gamma \in \mathcal{R}_X(\mathcal{A})} w_\gamma \ll (\log L)^{2e}. \quad (136)$$

Here $e = 0$ is the unweighted cell and $e = 1$ is the logarithmically weighted cell. Let

$$d\nu_{L,e}^{\text{ref}}(x, y) = \frac{e^x (\log L + x)^{2e-1}}{Z_{L,e}} dx e^y dy \quad (137)$$

be the finite- L reference probability measure of TPC-51. For a family $H = (H_\alpha)_\alpha$, with values in an arbitrary Hilbert space $\mathcal{K}_X$, define the normalized forms

$$\mathfrak{q}_X^0(H) := \frac{1}{E_X} \sum_{\alpha} \sum_{\gamma \in \mathcal{R}_X(\mathcal{A})} w_\gamma \|H_\alpha(z_\gamma)\|_{\mathcal{K}_X}^2, \quad (138)$$

$$\mathfrak{q}_X^M(H) := \frac{1}{E_X} \sum_{\alpha} \sum_{\gamma \in \mathcal{R}_X(\mathcal{A})} M_X(\alpha, \gamma) w_\gamma \|H_\alpha(z_\gamma)\|_{\mathcal{K}_X}^2, \quad (139)$$

$$\mathfrak{q}_{L,e}^{\text{ref}}(H) := \sum_{\alpha} \int_{[0,T]^2} \|H_\alpha(x, y)\|_{\mathcal{K}_X}^2 d\nu_{L,e}^{\text{ref}}(x, y). \quad (140)$$

Assume the literal mask has complete-universe row defect at most ϵ_X . Since the fixed cell has positive density, this implies

$$\sup_{\alpha} \#\{\gamma \in \mathcal{R}_X(\mathcal{A}) : M_X(\alpha, \gamma) = 0\} \ll_{\mathcal{A}} \epsilon_X |\mathcal{R}_X(\mathcal{A})|. \quad (141)$$

Put

$$\Delta_X^{\text{mom}} := e^{-c\sqrt{\log L}} + D^{-1/2+\delta}, \quad 0 < \delta < \frac{1}{2}, \quad (142)$$

where $c > 0$ is the fixed-modulus prime-progression constant. The quantitative input proved in TPC-51 is that, for every fixed $A_0 > 0$, there is a fixed C_1 such that

$$\left| \int e^{i(\xi x + \eta y)} d\nu_X - \int e^{i(\xi x + \eta y)} d\nu_{L,e}^{\text{ref}} \right| \leq C_{A_0} (1 + |\xi| + |\eta|)^{C_1} \Delta_X^{\text{mom}} \quad (143)$$

uniformly for $|\xi| + |\eta| \leq C_T (\log X)^{A_0}$ [6]. Here ν_X is the normalized empirical measure associated with (138). This displayed estimate, rather than weak convergence alone, is the arithmetic hypothesis used for a growing bank.

The error below has three logically different terms. The first is a smooth approximation tail. The second is the arithmetic moment error of the retained bank. The third is the energy removed by the literal mask.

Theorem 7.2 (Sampled row-log compression ledger). *Fix $A_0 > 0$. There are constants $C, C_0 < \infty$, depending on the fixed row cell and A_0 but not on $X, K_F, \mathcal{K}_X$, such that the following holds whenever $1 \leq K_F \leq (\log X)^{A_0}$. Let $(H_\alpha)_\alpha$ have finite source support and put*

$$\mathfrak{N}(H)^2 := \sum_\alpha \|H_\alpha\|_{\mathcal{C}^{2,2}}^2, \quad P_{K_F} H := (\Pi_{K_F}^{\text{row}} H_\alpha)_\alpha. \quad (144)$$

Then

$$\mathfrak{q}_X^M(H - P_{K_F} H) \leq \mathfrak{q}_X^0(H - P_{K_F} H) \leq \frac{C}{(K_F + 1)^2} \mathfrak{N}(H)^2, \quad (145)$$

$$\left| \mathfrak{q}_X^M(P_{K_F} H) - \mathfrak{q}_{L,e}^{\text{ref}}(P_{K_F} H) \right| \leq C \left(\Delta_X^{\text{mom}}(K_F + 1)^{C_0} + \epsilon_X(K_F + 1)^2 \right) \mathfrak{N}(H)^2. \quad (146)$$

In particular, the total approximation-and-calibration ledger obeys

$$\begin{aligned} & \mathfrak{q}_X^M(H - P_{K_F} H) + \left| \mathfrak{q}_X^M(P_{K_F} H) - \mathfrak{q}_{L,e}^{\text{ref}}(P_{K_F} H) \right| \\ & \leq C \left((K_F + 1)^{-2} + \Delta_X^{\text{mom}}(K_F + 1)^{C_0} + \epsilon_X(K_F + 1)^2 \right) \mathfrak{N}(H)^2. \end{aligned} \quad (147)$$

The exponent C_0 is fixed. One may take $C_0 = C_1 + 2$, where C_1 is the exponent in the prime-squarefree Fourier-moment error.

Proof. The mask only deletes nonnegative terms, so $\mathfrak{q}_X^M(H - P_{K_F} H) \leq \mathfrak{q}_X^0(H - P_{K_F} H)$. The normalized row weights have bounded total mass. Apply (135) for each α , then sum. This proves (145).

Write

$$P_{K_F} H_\alpha(x, y) = \sum_{0 \leq k_1, k_2 \leq K_F} a_{\alpha, k_1, k_2} \chi_{k_1, k_2}(x, y), \quad a_{\alpha, k_1, k_2} \in \mathcal{K}_X.$$

Products of two tensor cosines are sums of $O(1)$ exponentials with frequencies $O_T(K_F)$. The fixed-modulus prime-squarefree moment estimate therefore makes every sampled Gram entry differ from its finite- L reference entry by at most

$$C(1 + K_F)^{C_1} \Delta_X^{\text{mom}}.$$

There are $(K_F + 1)^2$ row-log modes. The Schur bound gives the operator error $C \Delta_X^{\text{mom}}(K_F + 1)^{C_1+2}$. This argument is unchanged for Hilbert-valued coefficients because the scalar Gram is tensored with $I_{\mathcal{K}_X}$.

For the mask term, the tensor-cosine Nikolskii estimate gives

$$\|P_{K_F} H_\alpha\|_\infty^2 \leq C_T(K_F + 1)^2 \sum_{k_1, k_2} \|a_{\alpha, k_1, k_2}\|_{\mathcal{K}_X}^2.$$

At most an ϵ_X -fraction of the complete opposite-row universe is removed for each α . The base-weight flatness on the fixed row cell therefore bounds the deleted bank energy by $C \epsilon_X(K_F + 1)^2 \sum_k \|a_{\alpha, k}\|^2$. Sum in α . Finally, Parseval and the definition of $\mathfrak{N}(H)$ bound the bank coefficient energy by $C \mathfrak{N}(H)^2$. Combining the arithmetic and deletion errors proves (146) and hence (147). $\square$

Proposition 7.3 (The uniformly active actual field meets the normalization). *Assume the compact C^∞ profile family of (22), the carrier bounds (84), and Assumptions 5.1 and 6.1. Let $\mathcal{U}_X$ be a finite set of source/cofactor pairs (α, r) , along the inherited regime $J, L \rightarrow \infty$. Let a_α be the actual source coefficient, independent of the retained cofactor label r , and let the source parameter be $\beta_\alpha \in \mathfrak{B}$. Define the retained field, indexed by the source label used in the literal mask, by*

$$H_{\alpha, X}(x, y) := \bigoplus_{r: (\alpha, r) \in \mathcal{U}_X} a_\alpha \bigoplus_{a \in A_0} \bigoplus_{j \in \mathcal{J}_{X, a}} F_{\beta_\alpha, a}(x, y, j/J) e_{r, a, j}. \quad (148)$$

The coordinate $e_{r,a,j}$ may be tensored with any retained residual basis vector; no coherent sum over those labels is taken. Put

$$\mathcal{C}_X^{\text{act}} := J \sum_{(\alpha,r) \in \mathcal{U}_X} |a_\alpha|^2 \mathcal{B}(\beta_\alpha). \quad (149)$$

Then, uniformly in the coefficients and parameters,

$$\mathfrak{N}(H_X)^2 \leq C_{\text{reg}} \mathcal{C}_X^{\text{act}}, \quad (150)$$

$$\mathfrak{q}_{L,e}^{\text{ref}}(H_X) \geq (\rho_* + o(1)) \mathcal{C}_X^{\text{act}}. \quad (151)$$

Consequently, if $K_{F,X} \rightarrow \infty$, then for all sufficiently large X ,

$$\mathfrak{q}_{L,e}^{\text{ref}}(P_{K_{F,X}} H_X) \geq c_{\text{ov}} \mathcal{C}_X^{\text{act}} \quad (152)$$

with a fixed $c_{\text{ov}} > 0$. Thus the two normalization hypotheses in (153) hold for the actual retained field, not merely for a model family.

Proof. For every derivative occurring in (133), orthogonality of the orbit coordinates gives

$$\sup_{x,y} \|\partial_x^v \partial_y^w H_{\alpha,X}(x,y)\|^2 \leq C J \sum_{r:(\alpha,r) \in \mathcal{U}_X} |a_\alpha|^2.$$

There are only finitely many derivative orders and direct cells. Sum in α , use Cauchy–Schwarz over those derivative orders, and then use $\mathcal{B} \geq b_-$ to obtain (150).

The progression Riemann sum in j/J , uniformly for the compact profile family, and uniform convergence of the finite- L reference density to e^{x+y} give

$$\mathfrak{q}_{L,e}^{\text{ref}}(H_X) = J \sum_{(\alpha,r) \in \mathcal{U}_X} |a_\alpha|^2 (\mathcal{P}(\beta_\alpha) + o(1)).$$

The error is uniform in α and β_α ; after weighting and summing, it is $o(\mathcal{C}_X^{\text{act}})$ because $b_- \leq \mathcal{B} \leq b_+$. Apply $\mathcal{P}(\beta_\alpha) \geq \rho_* \mathcal{B}(\beta_\alpha)$ to prove (151).

Finally, the weighted continuum estimate (134) and (150) imply

$$\|H_X - P_{K_{F,X}} H_X\|_{L^2(d\nu_{L,e}^{\text{ref}}; \mathcal{K}_X)} \leq C(K_{F,X} + 1)^{-1} (\mathcal{C}_X^{\text{act}})^{1/2}.$$

The triangle inequality, followed by $K_{F,X} \rightarrow \infty$, gives (152). $\square$

Corollary 7.4 (Polylogarithmic relative tail). *Suppose a retained actual family satisfies, for one declared normalization $\mathcal{C}_X > 0$,*

$$\mathfrak{N}(H_X)^2 \leq C_{\text{reg}} \mathcal{C}_X, \quad \mathfrak{q}_{L,e}^{\text{ref}}(P_{K_F} H_X) \geq c_{\text{ov}} \mathcal{C}_X \quad (153)$$

with fixed $C_{\text{reg}} < \infty$ and $c_{\text{ov}} > 0$. Let $K_F = \lfloor (\log X)^A \rfloor$, where $A > 0$ is fixed, and apply theorem 7.2 with a fixed $A_0 \geq A$. If

$$\Delta_X^{\text{mom}} K_F^{C_0} = o(1), \quad \epsilon_X (K_F + 1)^2 = o(1), \quad (154)$$

then $P_{K_F} H_X$ has masked energy comparable to $\mathcal{C}_X$, while

$$\mathfrak{q}_X^M(H_X - P_{K_F} H_X) \leq \mathfrak{q}_X^0(H_X - P_{K_F} H_X) \ll (\log X)^{-2A} \mathcal{C}_X = o(\mathcal{C}_X). \quad (155)$$

Thus the retained bank verifies the relative-tail alternative for this actual family.

If $L, D \geq X^{c_}$ for some fixed $c_* > 0$, $0 < \delta < 1/2$, and $\epsilon_X \leq X^{-\eta+o(1)}$ for some fixed $\eta > 0$, conditions (154) hold for every fixed $A > 0$.*

Proof. Insert $K_F = (\log X)^A$ in (147). The exponential prime error absorbs every fixed log power, the squarefree term has a fixed power saving because $\delta < 1/2$, and the mask term has a fixed power saving by hypothesis. The lower bound in (153) absorbs the resulting $o(\mathcal{C}_X)$ calibration error. Equation (155) is (145). $\square$

By proposition 7.3, this corollary applies to the uniformly active provenance-fixed field with $\mathcal{C}_X = \mathcal{C}_X^{\text{act}}$. Hence the relative-tail conclusion used in the introduction is an actual retained-label statement under the same exact-dictionary and profile-nondegeneracy hypotheses as the diagonal bridge.

7.3 What is and is not compressed

For the actual coefficient field, take

$$\mathcal{K}_X = \bigoplus_{r,j} \mathcal{K}_{X,r,j}, \quad (156)$$

where the summands retain the cofactor, orbit, and residual data required by (10); the source remains the family index, while the opposite-row coordinates (x, y) are the arguments of the field. The operator used above is exactly

$$\Pi_{K_F}^{\text{row}} \otimes I_{\mathcal{K}_X}. \quad (157)$$

It has $(K_F + 1)^2$ row-log modes, but every coefficient of those modes is a full vector in $\mathcal{K}_X$. In particular,

$$\dim \text{ran}(\Pi_{K_F}^{\text{row}} \otimes I_{\mathcal{K}_X}) = (K_F + 1)^2 \dim \mathcal{K}_X \quad (158)$$

when $\mathcal{K}_X$ is finite-dimensional. If the physical orbit rank is of order J , that factor is still present. Hence theorem 7.2 is compatible with scalar translate saturation: it compresses smooth dependence on the two row logs and makes no fixed-rank assertion about the orbit direction.

8 Support grazing: a sharp actual-direction obstruction

The positive profile-overlap hypothesis cannot be replaced by bounded smoothness, common compact support, or dyadic comparability of the scale parameters. We now give a fixed-shape example in which the carrier remains macroscopically nonzero while the physical profile collapses faster than every power. The construction concerns the provenance-fixed actual direction itself; it is not obtained by choosing an arbitrary Mellin coefficient density.

8.1 Two fixed flat bumps

Let

$$\vartheta(u) := \begin{cases} \exp(-1/u), & u > 0, \\ 0, & u \leq 0, \end{cases} \quad (159)$$

and fix the two nonzero shapes

$$U(t) := \vartheta(t-1)\vartheta(2-t), \quad V(t) := \vartheta(t-1)\vartheta(2-t). \quad (160)$$

Both functions belong to $C_c^\infty(\mathbb{R})$, are supported on $[1, 2]$, are positive on $(1, 2)$, and are flat at both endpoints. In particular, they belong to $C_c^\infty((0, \infty))$, as required by real-axis Mellin inversion. For $1 < c < 2$, define the carrier and the physical profile by

$$R_c(t) := U(t), \quad F_c(t) := U(t)V(ct). \quad (161)$$

The carrier does not depend on c . The two supports overlap only on

$$1 < t < \frac{2}{c}, \quad \frac{2}{c} - 1 = \frac{2-c}{c}. \quad (162)$$

Proposition 8.1 (Every-power overlap collapse). *Put*

$$\rho(c) := \frac{\int_{\mathbb{R}} |F_c(t)|^2 dt}{\int_{\mathbb{R}} |R_c(t)|^2 dt}. \quad (163)$$

For every $N > 0$, there is $C_N < \infty$ such that

$$0 \leq \rho(c) \leq C_N(2-c)^N \quad (1 < c < 2). \quad (164)$$

More precisely, there are absolute constants $C, c_0 > 0$ such that

$$\rho(c) \leq C(2-c) \exp\left(-\frac{c_0}{2-c}\right). \quad (165)$$

Proof. Write $\delta = 2 - c$ and $t = 1 + s$. On the overlap interval, $0 < s < \delta/c$, and

$$2 - ct = \delta - cs. \quad (166)$$

All factors in (160) are at most one. Using only the two flat factors facing the grazing endpoints gives

$$|U(1+s)V(c(1+s))|^2 \leq \exp\left(-\frac{2}{s} - \frac{2}{\delta - cs}\right).$$

Since $s < \delta/c$ and $0 < \delta - cs < \delta$, the right side is at most $\exp(-c_0/\delta)$ for an absolute $c_0 > 0$. The interval has length δ/c . The denominator in (163) is one fixed positive number, so (165) follows. Finally, $e^{-c_0/\delta} \ll_N \delta^N$ for every N , which proves (164). $\square$

Taking $c_X = 2 - X^{-\sigma}$, with fixed $\sigma > 0$, gives

$$\rho(c_X) \ll_M X^{-M} \quad \text{for every fixed } M > 0. \quad (167)$$

The shapes U, V have not changed with X ; only the legitimate scale parameter c_X approaches the grazing boundary.

8.2 The distinguished Mellin direction becomes a near-kernel

Let $\tilde{V}$ be the real-axis Mellin transform of V , and write

$$A_B(V) := \frac{1}{2\pi} \int_{\mathbb{R}} (1 + |\tau|)^B |\tilde{V}(\tau)| d\tau. \quad (168)$$

This is a fixed positive finite number. On positive samples $t_1, \dots, t_N$, take the carrier coefficients $b_n = U(t_n)$ and the physical samples $x_n = ct_n$. The canonical Mellin direction fixed by V is

$$\frac{1}{2\pi} \int_{\mathbb{R}} \tilde{V}(\tau) (b_n x_n^{i\tau})_{n=1}^N d\tau = (U(t_n)V(ct_n))_{n=1}^N. \quad (169)$$

Its envelope and output obey the exact ledger

$$\mathcal{E}_B^{\text{gr}}(c) = A_B(V) \left(\sum_n |U(t_n)|^2 \right)^{1/2}, \quad (170)$$

$$\|\mathcal{S}_V^{\text{gr}}(c)\|_2^2 = \sum_n |U(t_n)V(ct_n)|^2. \quad (171)$$

Thus support grazing shrinks the physical profile but not the carrier size seen by the canonical layer envelope.

The identical calculation on the Hilbert space $L^2((0, \infty), dt)$, with carrier $U(t)$, gives the continuum condition number

$$\frac{\mathcal{E}_B^{\text{gr}}(c)^2}{\|\mathcal{S}_V^{\text{gr}}(c)\|_2^2} = \frac{A_B(V)^2}{\rho(c)}. \quad (172)$$

Consequently (167) produces a condition number larger than every fixed power of X . This is a near-kernel for the one density $\tilde{V}$ prescribed by the physical shape. It is therefore different from the universal kernel obtained by allowing an arbitrary signed Mellin density.

8.3 Exact zero on a physical grid

The discrete obstruction can be stronger. Let $t_j = j/J$, and suppose the carrier window contains the grid points in $[1, 2]$. If

$$0 < \delta = 2 - c < \frac{c}{J}, \quad (173)$$

then

$$0 < \frac{2}{c} - 1 = \frac{\delta}{c} < \frac{1}{J}.$$

The open overlap interval $(1, 2/c)$ contains no grid point: $t = 1$ is an endpoint where $U(1) = 0$, and the next grid point is $1 + J^{-1} \geq 2/c$. Hence

$$U(t_j)V(ct_j) = 0 \quad \text{for every } j, \quad \sum_j |U(t_j)|^2 > 0 \quad (174)$$

for all sufficiently large J . The sampled physical output is then exactly zero although the canonical carrier envelope is positive. This example does not depend on an arithmetic sampling accident; it already occurs on the regular orbit grid. For example, $c_J = 2 - J^{-2}$ satisfies (173) for large J and gives both the exact grid zero and an every-power continuum overlap collapse.

8.4 A representation-free good–bad tradeoff

The correct energy selection is made on the unseparated physical profile, not on signed Mellin layers. Let R be a nonzero carrier, P a bounded physical multiplier, and

$$d\sigma := |R|^2 d\varpi, \quad \text{Bad}_\eta(P; R) := \frac{\sigma(\{|P| < \eta\})}{\sigma(\Omega)}, \quad \eta > 0. \quad (175)$$

Then the profile-to-carrier ratio satisfies

$$\frac{\int_\Omega |R|^2 |P|^2 d\varpi}{\int_\Omega |R|^2 d\varpi} \geq \eta^2 (1 - \text{Bad}_\eta(P; R)). \quad (176)$$

Conversely,

$$\frac{\int_\Omega |R|^2 |P|^2 d\varpi}{\int_\Omega |R|^2 d\varpi} \leq \eta^2 + \|P\|_\infty^2 \frac{\sigma(\{|P| \geq \eta\})}{\sigma(\Omega)}. \quad (177)$$

These inequalities are immediate by splitting the integral into the good and bad sets. They show exactly what an energy-selected boundary argument would need: a level $\eta = \eta_X$ for which the good set retains enough carrier mass and η_X itself does not introduce a forbidden polynomial loss. In the grazing example, the good carrier mass tends to zero for every fixed $\eta > 0$, and the best useful η_X is super-polynomially small.

The split in (175) is intrinsic to the physical multiplier P . It neither chooses a favorable term from a signed Mellin representation nor changes the distinguished coefficient density.

8.5 Scope of the obstruction

The construction proves sharpness of the positive-overlap assumption: uniform C^B bounds and compact scale control alone do not yield a uniform actual-direction condition number. It does *not* assert that the inherited TPC packet approaches this grazing configuration. That is decided only by inspecting its declared fixed cutoffs and normalized scale cell. Nor does the example involve coherent prime summation, residual grouping, localization to h_0 , a parity-sensitive correlation, or a prime-pair conclusion. It is a sharp analytic obstruction at the coefficientwise pre-terminal Mellin interface and nothing beyond that interface.

9 Endpoint ledger and claim boundary

This section records exactly what the preceding theorems purchase in the fixed-shift reduction. It is deliberately stated as an operator ledger, not as a prediction about primes.

9.1 Condition-number expenditure

For one direct cell and a sequence of nonzero coefficient vectors c_X , put $\Theta_X := \Theta(\mathbf{W}; c_X)$. The diagonal identity of [theorem 4.1](#) then assigns the scale-dependent cost

$$\kappa_{\text{can},B}(X) = \frac{A_B(\mathbf{W})^2}{\Theta_X}. \quad (178)$$

The Mellin factor $A_B(\mathbf{W})$ is fixed once the shapes, the number of factors, and B are fixed. Consequently:

$$\Theta_X \gg 1 \quad \implies \quad \kappa_{\text{can},B}(X) = O(1), \quad (179)$$

$$\Theta_X \geq X^{-o(1)} \quad \implies \quad \kappa_{\text{can},B}(X) = X^{o(1)}, \quad (180)$$

$$\Theta_X \geq X^{-\lambda+o(1)} \quad \implies \quad \kappa_{\text{can},B}(X) \ll X^{\lambda+o(1)}. \quad (181)$$

Here $\kappa_{\text{can},B}$ is a squared norm ratio. No square root is silently inserted into the exponent ledger.

The inherited endpoint reserves at most $1/400$ of polynomial exponent for this interface. The formal exponent ledger therefore requires

$$\lambda \leq \frac{1}{400}. \quad (182)$$

In an application carrying unspecified $X^{o(1)}$ factors, operational viability uses the strict inequality $\lambda < 1/400$; equality is a boundary case requiring a separate sharp endpoint estimate. Uniform activity has $\lambda = 0$. By contrast, the grazing example in [section 8](#) violates every fixed polynomial budget. These statements concern only this coefficientwise interface; unused budget here cannot be counted as progress at a downstream gate.

9.2 The precise forward step

The logical dependency is

$$\begin{array}{c} \text{TPC-50: a fixed-complexity direct cell survives the literal mask} \\ \Downarrow \\ \text{TPC-51: fixed scalar quotient frames and sampled row-log banks} \\ \Downarrow \\ \boxed{\text{TPC-52: a conditional pre-terminal Mellin criterion}} \\ \boxed{\text{on exact-dictionary, complete-row, uniformly active cells.}} \end{array} \quad (183)$$

The new point is that the physical orbit label is retained. The false scalar Gram is allowed to saturate, while the actual coefficient vector is treated by a diagonal multiplier and a nonnegative overlap ratio. Under the exact-profile dictionary, complete-row hypotheses (S1)–(S3), and $\rho_* > 0$, this verifies the actual-direction part of the narrow TPC-51 bridge and its row-log relative-tail branch on the declared retained field. Those hypotheses are not proved here for every inherited packet. The result is therefore a conditional L1 interface theorem and a finite, checkable reduction of the remaining provenance work.

9.3 Operations not licensed by the theorem

The following arrows remain absent:

- (O1) a lower square after the terminal coherent sum over primes p ;
- (O2) stability after identifying equal residual labels $s = d(m)r$;
- (O3) passage from the frozen shift family to the single physical shift h_0 ;
- (O4) identification of the coefficientwise quadratic form with the native affine physical Gram;
- (O5) reverse Minkowski control for the full signed projective profile used in TPC-48; and
- (O6) an estimate for any fixed-shift prime-pair correlation.

The diagonal proof would be invalid if applied after any operation that creates cross terms among (m, r, j, γ) . In particular, it cannot be moved through the coherent prime map merely by citing positivity of the pre-terminal energy.

9.4 Claim table

Level	Established here	Explicitly not established
L0	Scalar cluster saturation; orthogonal lift; exact multi-factor Mellin ledger; support-grazing obstruction.	A lower frame for every signed Mellin density.
L1	Conditional actual-direction stability and polylogarithmic row-log tail on exact-dictionary, complete-row, profile-nondegenerate direct cells.	Verification of those hypotheses for every inherited packet; coherent prime lower square, residual grouping, physical-shift localization, and native-Gram comparison.
L2	Nothing.	Hardy–Littlewood asymptotics, a parity-barrier breach, or any twin-prime conclusion.

Thus TPC-52 closes the analytic part of a declared interior subgate conditional on a finite actual-profile dictionary and sharpens the next decision. One must first verify that dictionary and uniform overlap for the packet in use; one may then test whether the later coherent-prime operator preserves enough of this retained-label energy, or prove that it does not. Boundary packets with vanishing overlap must be budgeted, discarded with a quantified physical error, or treated by additional structure; they are not covered by the interior theorem.

10 Exact finite certificate

The analytic arguments do not depend on floating-point experiments. The accompanying standard-library Python program

`experiments/tpc52_certificate.py`

is a deterministic regression test for their finite algebra. It uses exact rational arithmetic and performs 4,372 assertions in six groups:

- (C1) 3,612 clustered finite-difference moment identities;
- (C2) 128 orthogonal orbit-lift energy identities;
- (C3) 370 finite diagonal-multiplier overlap ledgers;
- (C4) 160 arbitrary nonnegative weight-transfer checks;
- (C5) 99 good/bad overlap threshold inequalities; and
- (C6) three exact endpoint fraction identities.

The normalized source SHA-256 is

3d6244dd53736436539c4842b732731919539164d001d1f41e8f886d80702ee4,

the semantic payload SHA-256 is

0fc0c178543beddd5ba86120d235fd28cd135ffdd84893cdda12cd296ca54bf9,

and the certificate-core SHA-256 is

2e136caf81b9ce92f8dcd55d834c78cf4eedc5c559d283f483e5d333177065.

The committed output is `experiments/tpc52_certificate_output.txt`. Running

```
python experiments/tpc52_certificate.py -output
experiments/tpc52_certificate_output.txt
```

reproduces it byte for byte when the normalized source is unchanged.

The certificate catches sign errors in the moment-annihilating coefficients, accidental reintroduction of scalar cross terms, inversion of the overlap ratio, failure of fiberwise lower bounds under arbitrary nonnegative concentration, and arithmetic mistakes in the endpoint ledger. It does not prove uniform prime-squarefree sampling, Sobolev tail estimates, flat-boundary asymptotics, or the physical profile nondegeneracy assumption. It does not enumerate a coherent prime sum and cannot certify a fixed-shift prime-pair statement. Its scope is finite L0 algebra supporting the stated L1 interface, with no L2 or parity-sensitive conclusion.

References

- [1] Liang Wang. Zero before separation in a primitive Möbius tail: Provenance-weighted closure and the principal row-mode boundary. Preprint, repository commit 1cf8d0da027c, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/1cf8d0da027cb2212251719a5396e6fcd43f0224/papers/tpc-25-zero-before-separation.
- [2] Liang Wang. Soft projective reassembly of the physical row mask: Orthogonal GCD compression, optimal Ferrers separation, and coherent product-slope spectra. Preprint, repository commit ecbd29afd77b, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/ecbd29afd77b92224aa7720c11c65444407fc0e9/papers/tpc-36-soft-projective-mask-reassembly.
- [3] Liang Wang. Orbit-band projection and mesoscopic spectral transfer for an actual-coefficient determinant sector. Preprint, repository commit dff4e13725fd, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/dff4e13725fddffe56f7eb63d6651da23978a7ba/papers/tpc-47-orbit-band-spectral-transfer.
- [4] Liang Wang. Critical orbit tiling and an all-band large-sieve reduction for the actual-coefficient residual determinant family. Preprint, repository commit f6dfd8764665, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/f6dfd8764665e60ac119af8aaffb64cffaf352dc/papers/tpc-48-critical-orbit-tiling-large-sieve.

- [5] Liang Wang. Fixed-complexity cell survival and sharp concentration obstructions for the TPC static mask. Preprint, repository commit 8fb4bef9aa0c, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/8fb4bef9aa0ce6d3e709015afaef00202eb18118/papers/tpc-50-fixed-complexity-cell-survival.
- [6] Liang Wang. Canonical mellin quotient frames: Finite-sampling kernels, interior stability, and the boundary of signed TPC reassembly. Preprint, repository commit 481243b, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/481243b/papers/tpc-51-canonical-mellin-quotient-frames.